

BIOLOGY GRADE 10: PLANT NUTRITION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Biology
Strand	Strand 2.0: Physiology of Plants and Animals
Sub-Strand	Sub-Strand 2.1: Plant Nutrition
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Autotrophic and heterotrophic nutrition in plants – Structure and function of the leaf as an organ of photosynthesis – Raw materials for photosynthesis: carbon dioxide, water, light energy, chlorophyll – Light-dependent and light-independent (dark) reactions of photosynthesis – Factors affecting the rate of photosynthesis: light intensity, carbon dioxide concentration, temperature, water availability – Experiments to investigate factors affecting the rate of photosynthesis (e.g., DCPIP decolourisation, oxygen bubble counting in Elodea/aquatic plants) – Leaf adaptations for photosynthesis (thin lamina, large surface area, stomata, chloroplasts, vascular bundles) – Mineral nutrition in plants: macronutrients and micronutrients, deficiency symptoms – Importance of photosynthesis to ecosystems and food chains (producer role) – Applications of plant nutrition: greenhouse farming, hydroponics, fertiliser use in Kenyan agriculture
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) distinguish between autotrophic and heterotrophic nutrition in plants, b) describe the structure of the leaf and relate it to its role as the primary organ of photosynthesis, c) explain the process of photosynthesis, including the light-dependent and light-independent reactions and the role of each raw material, d) investigate and explain the factors that affect the rate of photosynthesis through experiments, e) describe the role of mineral nutrients in plant growth and identify deficiency symptoms, f) evaluate the significance of photosynthesis to ecosystems, agriculture, and food security in Kenya, g) appreciate the importance of plant nutrition in sustaining life on Earth.
Core Competencies	– Communication and Collaboration: Learners collaboratively design and carry out experiments on photosynthesis (e.g., counting oxygen bubbles from aquatic plants in a Kisumu lake-water setup) and present findings to peers using annotated diagrams and data tables. – Critical Thinking and Problem Solving: Learners analyse experimental results on factors affecting photosynthesis, identify anomalies in data, and propose explanations for why sukuma wiki grown in a shaded kitchen garden produces smaller leaves than those grown in full sunlight. – Imagination and Creativity: Learners construct annotated 2D or 3D models of a leaf cross-section using locally available

	materials (cardboard, cellophane, drinking straws) to illustrate adaptations for photosynthesis. – Digital Literacy: Where devices are available, learners use simulations or videos to visualise the light-dependent and light-independent reactions and record observations in digital or written logs. – Self-efficacy: Learners develop confidence in scientific inquiry by designing fair-test investigations into one factor affecting the rate of photosynthesis and evaluating the reliability of their own results.
Core Values	– Responsibility: Learners handle laboratory chemicals (iodine solution, ethanol) and living specimens (leaves, aquatic plants) safely and dispose of waste appropriately, reflecting responsible stewardship of school resources and the environment. – Respect: Learners listen to and value peers' experimental observations and alternative explanations during group sensemaking discussions, appreciating diverse scientific thinking. – Unity: Learners work in mixed-ability groups during practical investigations, sharing roles equitably and recognising that collaborative effort produces more reliable scientific conclusions. – Social Justice: Learners connect plant nutrition to food security in Kenya, discussing how understanding photosynthesis and soil mineral nutrition can help smallholder farmers in the Rift Valley and Western Kenya improve crop yields.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners generate investigable questions from the anchoring phenomenon (e.g., "Why does the sukuma wiki wilt in the midday sun?") and refine them into testable hypotheses about water availability and photosynthesis rate. – Planning and Carrying Out Investigations: Learners plan and conduct fair-test experiments — including the variegated leaf starch test and the aquatic-plant oxygen-bubble experiment — controlling variables and selecting appropriate equipment. – Analysing and Interpreting Data: Learners tabulate, graph, and interpret data from photosynthesis rate experiments, identifying trends and explaining anomalous results using scientific reasoning. – Constructing Explanations: Learners build written and oral explanations linking leaf structure, raw materials, and environmental conditions to observed rates of photosynthesis, returning to the anchoring phenomenon at each stage. – Developing and Using Models: Learners iteratively revise a class model of the leaf and the photosynthesis process on the Driving Question Board, adding new mechanism layers after each lesson. – Obtaining, Evaluating, and Communicating Information: Learners read secondary sources on Kenyan agricultural research (e.g., KALRO recommendations on fertiliser use) and evaluate the evidence for mineral nutrient deficiency symptoms in maize and beans.
Pertinent & Contemporary Issues (PCIs)	– Environmental Conservation: Learners examine how deforestation in the Mau Forest reduces the photosynthetic capacity of ecosystems and discuss the consequences for carbon cycling, rainfall patterns, and local food production. – Food and Nutrition Education: Learners connect the productivity of photosynthesis in staple crops (maize, beans, sukuma wiki) to Kenya's national food security goals, linking plant biology to human dietary needs. – Agriculture and Food Security: Learners investigate how the use of fertilisers supplying macronutrients (N, P, K) and micronutrients on small-scale farms in the Rift Valley can correct mineral deficiencies and increase crop photosynthetic output. – Safety and Security: Learners follow laboratory safety protocols when using ethanol (flammable) during the starch test for photosynthesis experiment and when handling iodine solution, demonstrating responsible laboratory behaviour. – Financial Literacy: Learners evaluate the cost-benefit of using inorganic fertilisers versus organic compost for a maize farmer in Nakuru, connecting understanding of mineral nutrition to practical economic decision-making.
Career Connections	– Botanist: This sub-strand provides foundational knowledge of plant physiology and photosynthesis that underpins degree-level botanical research, including work at institutions such as the National Museums of Kenya. – Agricultural Scientist / Agronomist: Understanding mineral nutrition, photosynthesis, and factors limiting plant growth is directly applied by agronomists advising Kenyan smallholder farmers on fertiliser use and crop management with organisations like KALRO.

	– Horticulturist / Greenhouse Farmer: Knowledge of optimising light intensity, CO₂ concentration, and mineral nutrition is essential for running commercial greenhouse operations (e.g., flower and vegetable farms in Naivasha and Thika). – Environmental Scientist: Understanding photosynthesis as the engine of primary production and carbon fixation connects to careers in climate science, carbon sequestration, and ecosystem monitoring in Kenya's forests and wetlands. – Biochemist / Biotechnologist: The light-dependent and light-independent reactions introduce metabolic pathways (ATP synthesis, Calvin cycle) that form the basis of biochemistry and biotechnology careers, including development of drought-resistant crops at KARI/KALRO. – Secondary School Biology Teacher: Mastery of plant nutrition and photosynthesis prepares learners to eventually communicate these concepts as educators, supporting Kenya's Vision 2030 science education goals.
Focus for Lessons	This unit anchors learning in a locally familiar, puzzling observation — sukuma wiki and spinach leaves wilting in the hot midday sun in a Nairobi kitchen garden — and uses it to drive a 10-lesson investigation into how plants make their own food through photosynthesis and how they obtain the mineral nutrients needed to grow. Learners move from initial predictions about why plants wilt and why leaves are green, through hands-on experiments and data analysis, toward a mechanistic understanding of photosynthesis (including light-dependent and light-independent reactions), the role of leaf structure, and the importance of mineral nutrition. Throughout the unit, every new concept is connected back to the anchoring phenomenon and to the real-world context of Kenyan agriculture, food security, and environmental conservation.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does our sukuma wiki wilt in the midday sun even when the soil is wet — and what does a plant actually need to stay green, grow, and make its own food? KICD KEY INQUIRY QUESTIONS: 1. How do plants make their own food? 2. What factors affect the rate at which plants make food?
Anchoring Phenomenon	A form 4 student in Nairobi notices that the sukuma wiki (kale) plants in her family's kitchen garden look healthy and upright in the early morning, but by midday — when the sun is strongest — the leaves have wilted visibly, even though the soil appears moist. The following morning they have recovered. The student also notices that when the plants are moved under the shade of a mango tree, the wilting stops but the leaves gradually become pale yellowish-green over several weeks and the plant grows more slowly. This phenomenon is puzzling because: (1) wilting happens even when water seems available in the soil, suggesting something about the rate of water movement or use is changing; (2) shade prevents wilting but causes a new problem — pallor and slow growth — suggesting that light is essential for something the plant needs to do; and (3) the plant recovers overnight, implying a reversible process rather than permanent damage. These observations create genuine puzzlement that motivates questions about how plants get energy, what role light plays, why leaves are green, and what the plant needs from the soil and air to stay healthy and grow.
Storyline Thread	Lesson 1 — ANCHORING PHENOMENON & PREDICT: Students observe photographs and a short description of the wilting sukuma wiki phenomenon. They write individual predictions about why it wilts and why shade causes pallor, and contribute opening questions to the class Driving Question Board (DQB). The teacher introduces autotrophic vs. heterotrophic nutrition to frame the unit question. Lesson 2 — LEAF STRUCTURE & OBSERVE: Students dissect or examine prepared cross-sections of a dicot leaf (e.g., from a locally available plant such as a mango or bean leaf) under a hand lens or microscope. They identify and annotate the epidermis, palisade mesophyll, spongy mesophyll, stomata, guard cells, and vascular bundles, linking each feature to its function in photosynthesis and water movement — beginning to explain the wilting phenomenon.

	Lesson 3 — RAW MATERIALS & EXPLAIN: Through a structured reading and guided discussion, learners identify the four raw materials for photosynthesis (CO₂, H₂O, light energy, chlorophyll) and write the word equation and chemical equation. They revisit the DQB: which raw materials could explain wilting (water/light) and pallor (light/chlorophyll)? Lesson 4 — STARCH TEST PRACTICAL (OBSERVE): Learners carry out the standard starch test on a variegated leaf (or a leaf that has been partially covered with black paper) using ethanol decolourisation and iodine solution. They observe that only the green, illuminated sections turn blue-black, providing evidence that chlorophyll and light are both required for photosynthesis to produce starch. DQB is updated. Lesson 5 — LIGHT-DEPENDENT REACTIONS (EXPLAIN): Using a diagram and guided notes, learners trace the light-dependent reactions: absorption of light by chlorophyll, photolysis of water, production of ATP and NADPH, and release of oxygen. They connect this to the oxygen bubbles observed in the supporting phenomenon and to the role of light in the sukuma wiki phenomenon. Lesson 6 — LIGHT-INDEPENDENT REACTIONS / CALVIN CYCLE (EXPLAIN): Learners explore the light-independent reactions: fixation of CO₂ using ATP and NADPH from the light stage, production of glucose (G3P). They construct a simplified flow diagram linking both stages and return to the DQB to explain why shaded sukuma wiki grows slowly (less ATP and glucose produced). Lesson 7 — FACTORS AFFECTING RATE OF PHOTOSYNTHESIS — INVESTIGATION (PLAN & OBSERVE): Learners plan and carry out a quantitative investigation using an aquatic plant (or a simulated dataset if live plants are unavailable), counting oxygen bubbles per minute at different light intensities (covering the lamp with layers of muslin or changing distance). They record data in tables and plot line graphs, identifying the limiting factor concept. Lesson 8 — FACTORS AFFECTING RATE — DATA ANALYSIS & EXPLAIN: Learners analyse their graphs from Lesson 7 and examine provided datasets for the effects of CO₂ concentration and temperature on photosynthesis rate. They apply the concept of limiting factors to explain real agricultural scenarios (e.g., why greenhouse tomato farmers in Naivasha add CO₂ to boost yields). Lesson 9 — MINERAL NUTRITION (OBSERVE & EXPLAIN): Using photographs and secondary data from KALRO research, learners identify macronutrients (N, P, K, Ca, Mg, S) and micronutrients, describe their roles in plant growth, and match deficiency symptoms to specific minerals (e.g., yellowing of older leaves = nitrogen deficiency in the Nakuru maize farmer supporting phenomenon). They connect mineral uptake via roots to the overall nutrition of the sukuma wiki plant. Lesson 10 — MODEL BUILDING, SYNTHESIS & DQB RESOLUTION: Learners produce a final annotated model (poster or digital diagram) that integrates leaf structure, the two stages of photosynthesis, limiting factors, and mineral nutrition to fully explain the original anchoring phenomenon (why sukuma wiki wilts at midday and turns pale in shade). Groups present their models, the class resolves all remaining DQB questions, and learners write a final evidence-based explanation connecting plant nutrition to Kenyan agriculture and food security.
Supporting Phenomena	– A maize farmer in Nakuru notices that one section of her shamba has stunted, yellow-leaved maize plants despite receiving the same rainfall as the healthy green section — prompting questions about what else plants need from their environment besides water and light (mineral nutrition / nitrogen deficiency). – A student conducting a DCPIP decolourisation experiment in the school lab observes that the blue dye fades faster when a test tube of leaf extract is placed near a bright window than when kept in a dark cupboard — raising questions about the role of light energy in chemical reactions inside leaves.

	– Aquatic plants (e.g., Elodea or locally available pond plants from Lake Victoria) visibly produce more oxygen bubbles per minute when placed under a bright lamp than under dim light or in the dark — providing a quantitative, observable proxy for the rate of photosynthesis. – Variegated leaves (green and white sections) of a commonly available ornamental plant, when tested for starch with iodine solution after a period in the light, show blue-black colouration only in the green sections — demonstrating that chlorophyll is required for photosynthesis to produce starch.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (40 minutes): Why Does Our Sukuma Wiki Wilt? Anchoring the Phenomenon and Opening Our Questions

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To anchor learners in a puzzling, familiar Kenyan agricultural phenomenon that will drive inquiry across the entire sub-strand on plant nutrition
Knowledge	Learners will distinguish between autotrophic and heterotrophic nutrition and state that green plants make their own food using light energy
Skills	Learners will record observations, write evidence-based predictions, and generate investigable questions about the sukuma wiki phenomenon
Attitudes	Learners will demonstrate curiosity and open-mindedness by sharing predictions without fear of being wrong, and appreciate that everyday observations in Kenyan homes and gardens can be the starting point for scientific inquiry
Key Inquiry Question	How do plants make their own food?
Purpose in Storyline	This is the opening anchoring lesson. It launches the phenomenon that every subsequent lesson will return to and explain more fully. By the end of this lesson learners have a shared puzzlement, a public record of their initial ideas, and a Driving Question Board that will grow across all 10 lessons until it is fully resolved in Lesson 10
Safety Notes	No chemicals or sharp instruments are used in this lesson. If printed photographs are handled, ensure edges are not sharp. If learners are asked to bring a sukuma wiki leaf from home, remind them to wash hands after handling any plant material.

B. LESSON OVERVIEW

Learners are introduced to the central anchoring phenomenon: a Nairobi student observes that her family sukuma wiki wilts at midday despite moist soil, recovers overnight, but turns pale and grows slowly when moved into shade. Through individual written predictions, a think-pair-share discussion, and a class Driving Question Board, learners surface prior knowledge and genuine puzzlement. The teacher then introduces the key distinction between autotrophic and heterotrophic nutrition to frame the unit. The lesson closes with learners identifying at least two questions they personally want answered, linking to the driving question that will guide the whole sub-strand.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Photosynthesis in organisms <i>Related topic: not an exact match for this lesson.</i> Source: Khan	Learners are shown two high-quality photographs displayed on the board or printed on A4 sheets: Photograph A shows a healthy upright sukuma wiki plant in the early morning in a Nairobi kitchen garden, and Photograph B shows the same patch of sukuma wiki	The teacher displays the photographs and says: 'Look carefully at these two images of sukuma wiki — a plant most of you have eaten or seen growing at home. I want you to notice something puzzling. In the morning the plants look strong and green.'	Individual written prediction activates prior knowledge and creates a baseline record of each learner's initial conceptual model. Photographs ground the phenomenon in a familiar Kenyan context, reducing the cognitive distance between science class	Teacher reads over shoulders during silent writing to identify the range of prior ideas: do learners link wilting to water? Do any mention sunlight or temperature? Do any mention food-making? These observations inform which student ideas to amplify in the

Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: plant nutrition autotrophic heterotrophic	visibly wilted at midday with the soil appearing dark and moist. A third image shows the sukuma wiki after being moved under a mango tree: the leaves are now pale yellowish-green and the plant looks stunted. The teacher reads aloud a short narrative about Amina, a Form 4 student from Kasarani, Nairobi, who observes all three of these changes in her family garden. Learners are given 4 minutes of silent individual writing time to respond to three prompt questions in their notebooks: (1) Why do you think the sukuma wiki wilts at midday even though the soil is wet? (2) Why do you think it recovers by the next morning? (3) Why do you think moving it into shade causes the leaves to turn pale and the plant to grow slowly? Learners are explicitly told that there are no wrong answers at this stage and that their predictions will be revisited and refined throughout the unit.	By midday, with the soil still wet, the leaves are drooping. And when Amina moved the plants into the shade to stop the wilting, something new went wrong. Before we learn anything new, I want to know what YOU think is happening. Write your predictions in your notebook — full sentences please. I am not looking for the right answer. I am looking for YOUR thinking.' The teacher then circulates silently for 4 minutes, reading over shoulders without commenting, noting interesting predictions to return to later. WAIT TIME: Allow a full 4 minutes of silent individual writing before moving to pairs. Do not prompt or hint during this time.	and real life.	discussion. Teacher mentally notes two or three predictions that represent contrasting ideas to share with the class.
Observe Phase VIDEO: Photosynthesis in organisms <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos READING: <i>No closely matching reading in the ARES library for this activity.</i>	Learners share their predictions first in pairs for 2 minutes, then the teacher takes 4-5 predictions from different pairs and writes them verbatim on the board under the heading 'Our Current Thinking.' The teacher does not evaluate any prediction as right or wrong. Predictions might include: 'The sun heats the water and it dries up,' 'The plant uses water faster than the roots can take it in,' 'Shade is cold so the plant cannot make food,' or 'The plant needs sunlight to stay green.' After recording predictions, the teacher displays a third image: a photograph of a mushroom growing on a log next to a photograph of a succulent	After pairs have discussed, the teacher asks: 'Who heard something from their partner that was different from their own prediction — something that made you think?' Teacher calls on 3 learners to share. Then the teacher says: 'Look at these two organisms. Both are alive. Both need energy. But notice — the maize is out in the open sun, making something for itself. The mushroom is tucked under a log, getting energy from something that was once alive. What does that difference tell us about how they survive?' WAIT TIME: After posing the maize-mushroom question, wait 30 seconds in silence before	Paired discussion lowers the affective barrier to sharing ideas publicly. The maize-mushroom contrast is a locally recognisable juxtaposition that sets up the autotroph-heterotroph distinction without using the terms yet, allowing learners to construct the concept before receiving the vocabulary.	Listen to pair discussions: can learners articulate that the green plant uses sunlight to make food while the mushroom cannot? This reveals whether learners hold a naive view that plants eat from the soil the way animals eat food, which is a known misconception to address in Lesson 3.

 Search ARES for readings Search terms: plant nutrition autotrophic heterotrophic	green maize plant growing in a Rift Valley farm. The teacher asks learners to observe and discuss in pairs: 'What is the most important difference between how a mushroom gets its food and how a maize plant gets its food?' Pairs share responses. The teacher collects the key idea that the maize plant makes its own food while the mushroom depends on other organic matter, but does not yet formally define the terms.	taking responses. Allow learners to sit with the question. Do not rephrase or simplify it.		
Explain Phase  VIDEO: Photosynthesis in organisms <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: plant nutrition autotrophic heterotrophic	The teacher introduces the formal vocabulary of autotrophic and heterotrophic nutrition. Learners copy a structured two-column comparison into their notebooks: Autotrophic Nutrition (auto = self, trophe = feeding) — the organism manufactures its own organic food from simple inorganic raw materials using an external energy source such as light; examples include all green plants, algae, and some bacteria. Heterotrophic Nutrition — the organism cannot manufacture its own food and must consume ready-made organic molecules from other organisms; examples include all animals, fungi, and most bacteria. The teacher connects this directly to the phenomenon: 'Amina noticed that her sukuma wiki wilted and turned pale. If the sukuma wiki plant is autotrophic — if it makes its own food — what must it need to do that? And what happens to the plant if it cannot make its food properly?' Learners write one sentence in their notebooks completing this stem: 'If sukuma wiki is autotrophic, it must need _____ to make food, and if it cannot do this, the plant will _____.' The teacher takes 4-	The teacher writes AUTOTROPHIC and HETEROTROPHIC on the board, breaking down the Greek roots. Then says: 'These are not just vocabulary words. These words tell us something important about the sukuma wiki problem. If Amina is going to figure out why her plant is struggling, she needs to understand what it means for a plant to feed itself. Let us think about what that might require.' After learners write their sentence completions, the teacher reads two or three aloud and says: 'Hold onto these ideas. We are going to test them, challenge them, and improve them over the next few weeks. By Lesson 10 you will be able to answer this question with real evidence.' WAIT TIME: After asking learners to complete the sentence stem, give 2 full minutes of silent writing time before taking responses.	Introducing autotrophic and heterotrophic nutrition as a conceptual frame — rather than as isolated definitions — gives learners a lens through which to view all subsequent content. Connecting the terms immediately back to the sukuma wiki phenomenon prevents vocabulary from becoming inert knowledge.	Read the sentence completions. Key things to look for: do learners identify light as a requirement? Do any mention water, soil, or air? Do any connect failure to make food to wilting or pallor? These will confirm which raw materials are already in learners mental models, informing the emphasis needed in Lesson 3.

	5 responses verbatim and writes them on the board, noting that the class will spend the next several lessons finding out exactly which raw materials are needed and what each one does.			
Driving Question Board (DQB) Creation  VIDEO: Photosynthesis in organisms <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: plant nutrition autotrophic heterotrophic	Each learner receives two sticky notes (or cuts two small squares from scrap paper if sticky notes are unavailable). On the first note, they write one question they have about why the sukuma wiki wilts or turns pale — a question they genuinely do not yet know the answer to. On the second note, they write one question they have about how plants make food in general. Learners bring their notes to a large sheet of manila paper or a section of the classroom wall designated as the Driving Question Board. The teacher reads the driving question aloud from the top of the board: 'Why does our sukuma wiki wilt in the midday sun even when the soil is wet — and what does a plant actually need to stay green, grow, and make its own food?' Learners stick their questions below the driving question. The teacher then reads selected questions aloud and works with the class to group them loosely into emerging clusters: questions about water and wilting, questions about light and colour, questions about food-making, and questions about what plants take from soil and air. The clusters are labelled with card strips. Learners are told that the DQB will be revisited and updated after every lesson.	The teacher says: 'This board belongs to you. Every question up here was written by someone in this room. Over the next ten lessons we are going to answer these questions one by one using evidence — from experiments, from data, from what scientists in Kenya and around the world have discovered. At the end of the unit, every question on this board should have an answer — an answer YOU can explain.' The teacher reads 5-6 questions aloud, placing them on the board and asking the class: 'Does this question fit with these others about water, or does it belong with the questions about light?' to demonstrate the clustering process. Then learners help place the remaining questions. WAIT TIME: After asking 'What question are you most curious about from today?' pause 20 seconds before taking responses. Honour whatever question a learner shares, even if it goes beyond the syllabus.	The DQB makes learner curiosity visible and public, creating shared intellectual ownership of the inquiry. Grouping questions by theme previews the structure of the sub-strand and helps learners see that their genuine questions align with the key inquiry questions of the curriculum.	The questions on the DQB are a rich formative record. Teacher photographs the board or writes a list of all questions. Look for: How many learners ask about water versus light versus soil? Are there questions that reveal misconceptions (e.g., do plants breathe like animals)? Are there questions that anticipate later lessons (e.g., what is chlorophyll)? Use this record to prioritise examples and emphasis in Lessons 2 through 10.
Model Building Phase	Learners spend the final 5 minutes drawing an initial model in their notebooks. The prompt is: 'Draw a	The teacher says: 'Scientists do not just write notes — they build models. A model is a way of	The initial model externalises each learner existing conceptual model before instruction, creating a	Collect or photograph initial models to compare with Lesson 10 final models. Look for common

 VIDEO: Photosynthesis in organisms <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: plant nutrition autotrophic heterotrophic	simple sketch of Amina sukuma wiki plant and use arrows or labels to show what you currently think is going INTO the plant, what is HAPPENING INSIDE the plant, and what is COMING OUT of the plant. You can use words and pictures. This is your starting model — not your final answer.' Learners label their sketch with the date and write: 'My Starting Model — Lesson 1.' They are told they will add to, revise, and improve this model after each lesson, and that by Lesson 10 their final model will be detailed, accurate, and supported by evidence.	showing how you think something works. Your model right now will probably be incomplete and some parts might be wrong — and that is exactly right for Lesson 1. I want to see YOUR thinking, not a textbook diagram. We will return to this model in every lesson and watch it grow.' Teacher circulates, asking individual learners: 'What made you choose that arrow?' or 'What do you think is happening inside the leaf?' to prompt elaboration without correcting. WAIT TIME: Allow the full 5 minutes without interruption. Resist the urge to show a correct model or textbook diagram at this stage.	baseline for tracking conceptual change across the unit. The practice of revisiting and revising the model after each lesson builds metacognitive awareness of learning as a process of refinement rather than replacement.	starting points: do learners show soil going in (correct), sun shining on the plant (correct, partially), water being absorbed (correct), oxygen or food coming out (rare at this stage)? Note which learners already include light and which focus entirely on soil and water — this informs differentiation in Lesson 3.
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D. TEACHER REFLECTION

After the lesson, ask yourself: (1) Did learners engage with the phenomenon as genuinely puzzling, or did they seem to already know the answer? If the latter, probe more deeply in Lesson 2 about the mechanism, not just the outcome. (2) What misconceptions appeared in the predictions and DQB questions? Common ones include: plants eat from soil the way animals eat food, wilting is caused by heat drying the soil rather than by water movement within the plant, and shade is cold rather than low in light energy. Plan to address these explicitly in Lessons 3 and 5. (3) Were there learners who did not contribute to the DQB? Find a private moment before Lesson 2 to invite their questions individually. (4) Review the photograph of the DQB — are all four curriculum themes (water and wilting, light and colour, food-making processes, soil nutrients) represented in learner questions? If any cluster is thin, plan a teacher-posed question to seed it in Lesson 2. (5) Were predictions written in full sentences with reasoning? If not, reinforce the science writing expectation in the Lesson 2 notebook prompt.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed photographs showing sukuma wiki wilting at midday despite moist soil, recovering overnight, and turning pale when moved into shade under a mango tree in a Nairobi kitchen garden
What did I learn?	We learned that green plants are autotrophs — they manufacture their own food from simple inorganic raw materials — while organisms like mushrooms are heterotrophs that depend on ready-made organic food from other sources
How does this explain the phenomenon?	We cannot yet fully explain why the sukuma wiki wilts at midday or why it turns pale in shade, but we have recorded our starting predictions and questions on the Driving Question Board and will return to them with evidence in every lesson that follows

LESSON 2 (80 minutes): Leaf Structure: The Architecture of a Photosynthetic Machine

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To examine the internal and external structure of a dicot leaf and link each structural feature to its function in photosynthesis and water movement, advancing the class explanation of the sukuma wiki wilting phenomenon.
Knowledge	By the end of the lesson, the learner should be able to: (a) identify and label the structural features of a dicot leaf (epidermis, cuticle, palisade mesophyll, spongy mesophyll, stomata, guard cells, vascular bundles); (b) explain how each feature is adapted to its function in gas exchange, water transport, and light capture; (c) relate leaf structural features to the wilting and recovery pattern of the sukuma wiki plant.
Skills	By the end of the lesson, the learner should be able to: (a) prepare and observe a hand-lens or microscope specimen of a leaf cross-section; (b) produce a fully annotated scientific diagram of a dicot leaf cross-section; (c) construct evidence-based explanations linking structural features to the sukuma wiki phenomenon.
Attitudes	By the end of the lesson, the learner should be able to: (a) appreciate the elegant design of leaf structures as solutions to competing demands (light capture vs. water loss); (b) show curiosity and precision when handling biological specimens and microscopes; (c) value collaborative sense-making when interpreting observations.
Key Inquiry Question	How do plants make their own food?
Purpose in Storyline	In Lesson 1, students anchored their puzzlement in the sukuma wiki phenomenon and generated initial questions on the Driving Question Board. In this lesson, students gather the first direct structural evidence — the internal anatomy of a leaf — that begins to explain why wilting occurs even in moist soil (stomata open under bright light, driving rapid transpiration) and why shade causes pallor (reduced light reaching palisade cells). This evidence advances the class toward a mechanistic explanation of photosynthesis and water movement.
Safety Notes	Razor blades or scalpels used for cross-sections must be handled under direct teacher supervision. Students must not run when moving around lab benches. Microscope lenses should be cleaned only with lens tissue. Wash hands after handling plant material. If iodine solution is used as a staining aid, warn students it is an irritant — avoid contact with eyes and skin.

B. LESSON OVERVIEW

Students examine freshly cut or prepared cross-sections of a locally available dicot leaf (mango — *Mangifera indica* — or bean leaf) under a hand lens and, where available, a light microscope. Working in pairs, they produce annotated scientific diagrams identifying the upper and lower epidermis, cuticle, palisade mesophyll, spongy mesophyll, air spaces, stomata, guard cells, and vascular bundles (xylem and phloem). Through structured discussion, they link each layer's position and features to its function: the transparent cuticle and epidermis admit light; the densely packed palisade cells maximise chloroplast exposure; the air spaces in the spongy mesophyll facilitate CO₂ diffusion; the stomata regulate gas exchange and water vapour loss; the xylem supplies water and minerals while phloem exports sugars. Students then revisit the sukuma wiki phenomenon, using their structural evidence to begin explaining why stomata opening at midday drives water loss faster than xylem can supply it — initiating turgor loss and wilting — and why shade reduces the photosynthetic capacity of palisade cells, causing pallor and slow growth. Students update the class Driving Question Board (DQB) with new questions generated by today's evidence.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 12 Plant Transport Xylem And Phloem. Transpiration [3D Animation] <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  HTML: Leaf Structure and Function - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Each student receives a whole fresh mango or bean leaf and a hand lens. Without cutting it open, they sketch what they think the inside of the leaf looks like in cross-section — labelling any layers or structures they hypothesise exist and writing one sentence explaining why they think those layers are there. Students then share their prediction sketch with a shoulder partner and note one similarity and one difference between their predictions. The teacher collects 4–5 prediction sketches on the board using magnets (or Blu-Tack) so the class can see the range of ideas. Students are told explicitly: 'We will return to your predictions at the end of today's observation to see which parts were accurate and which we need to revise.'	Distribute leaves and hand lenses as students settle. Say: 'Before we cut anything open, I want you to think like a detective. Look at the outside of this leaf — feel it, hold it up to the light, look at both surfaces. Now sketch what you think it looks like on the INSIDE — like a slice through a loaf of githeri bread. You have 4 minutes. Go.' [Start timer. Circulate silently for 2 minutes, observing sketches without commenting.] After 2 minutes say: 'Now compare with your partner — find ONE thing you both drew and ONE thing that is different. 30 seconds.' [After 30 seconds] Cold-call 3 pairs: 'Amina and Baraka — what did you both draw?' [10-second wait time after each question before accepting answers.] Select 4–5 sketches that show a range of ideas — from single-layer to multi-layer predictions — and post them on the board. Say: 'Keep these predictions visible. Great scientists revise their ideas when evidence surprises them — let's see what the leaf actually looks like.'	Prediction Sketching with Peer Comparison: By committing a prediction to paper before observing, students activate prior knowledge and create a cognitive 'gap' between their mental model and reality. Posting predictions publicly creates accountability and models the scientific norm that initial models are expected to be revised. This directly mirrors how the sukuma wiki student in Nairobi was puzzled — she had a mental model (water in soil = healthy plant) that the evidence contradicted.	Teacher circulates and notes: (1) Do students draw any layers at all, or a single uniform interior? (2) Does any student spontaneously include stomata or veins? (3) Do prediction sketches reference water, light, or gas? This baseline reveals prior knowledge gaps to target during the Observe phase. Observable indicator: every student has a completed prediction sketch with at least one labelled layer before specimens are distributed.
Observe Phase  VIDEO: 12 Plant Transport Xylem And Phloem. Transpiration [3D Animation] <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  HTML:	Students work in pairs at lab benches. Each pair receives: a fresh mango or bean leaf, a pre-prepared microscope slide of a dicot leaf cross-section (or a prepared hand-section made by the teacher using a razor blade and two flat pieces of pith/carrot as guides), a hand lens, and — where microscopes are available — a light microscope set to 40× and 100×. Students follow a structured observation protocol printed on their activity card: Step 1 — Examine the whole leaf with a	Before observation begins, demonstrate the hand-sectioning technique at the front bench: 'Watch carefully — I am going to show you how to take a very thin slice. The thinner the slice, the more you can see.' [Demonstrate holding the leaf folded between two pieces of carrot, making a firm single stroke with the razor blade.] 'Now — safety first: the blade moves AWAY from your body, always. If you are unsure, raise your hand and I will help you.' Distribute materials. Say: 'You	Structured Observation Protocol with Scaffolded Diagram Annotation: The printed outline diagram reduces cognitive load so students focus on interpreting what they see rather than on drawing from scratch. The deliberate progression from hand lens → low magnification → high magnification builds the habit of moving from macroscopic to microscopic evidence — a core NGSS Science and Engineering Practice (Planning and Carrying Out Investigations). Projecting the	Teacher checks each pair's annotated diagram during circulation. Observable indicators: (1) At least 4 distinct layers are shaded and labelled by the end of Step 2. (2) Every student locates and sketches at least one stoma with guard cells. (3) Students are drawing what they actually see, not what they remember from a textbook (check for personal variation in sketches rather than identical copied diagrams). Red flag: a student who cannot distinguish the upper from lower

Leaf Structure and Function - Advanced (Flexbook) Source: CK-12 Search ARES for similar readings	hand lens: observe the upper vs. lower surface texture, shine, and vein pattern. Record observations using annotated sketches. Step 2 — Examine the cross-section under a hand lens first, then under the microscope (40×, then 100×). Step 3 — On the printed outline diagram provided, shade and label every distinct layer or region they can see, using pencil. Step 4 — At the microscope, locate stomata on the lower epidermis and sketch a single stoma with its two guard cells at high magnification. Students also examine a second slide or use a digital image displayed on the classroom screen showing a stained leaf cross-section so all students — including those without individual microscopes — can observe the full structure. Throughout, the teacher projects the anchor image of the wilting sukuma wiki plant and periodically prompts: 'As you observe each layer, ask yourself — could this layer be connected to why the sukuma wiki wilts?'	have 20 minutes to complete all four steps on your activity card. I am timing you. Step 1 should take about 4 minutes. Begin.' [Circulate continuously. At the 4-minute mark, prompt:] 'Move to Step 2 now.' [At the 10-minute mark, cold-call one pair:] 'Wanjiru and Omondi — tell the class one layer you can see and where it is.' [10-second wait time.] [At the 15-minute mark, say:] 'Everyone — find the stomata on the lower surface. They may look like tiny mouths or pores. Raise your hand when you have found one.' [Count hands. If fewer than half have found stomata, project the annotated diagram on screen and say:] 'Look at this region here — lower epidermis — scan slowly across this area.' At 20 minutes: 'Pencils down on observation. We will now make sense of what we saw.'	anchor phenomenon image during observation maintains the explanatory goal: students are not just cataloguing structures for their own sake but gathering evidence relevant to a real puzzling event.	epidermis — teacher intervenes with: 'Feel both surfaces of the leaf — which side is shinier? Why might that be? Look at which side has more stomata.'
Explain Phase  VIDEO: 12 Plant Transport Xylem And Phloem. Transpiration [3D Animation] <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria Search ARES for similar videos  HTML: Leaf Structure and Function - Advanced (Flexbook)	The class transitions from observation to structured sensemaking. The teacher projects a clean, unlabelled diagram of a dicot leaf cross-section on the board. Using a jigsaw-style expert group activity, each group of 4 is assigned one structural feature (epidermis/cuticle; palisade mesophyll; spongy mesophyll/air spaces; stomata/guard cells; vascular bundles). Each group has 6 minutes to discuss and prepare a 60-second 'function statement' — one sentence describing what their structure looks like, one sentence explaining what it does,	Transition announcement: 'Close your observation notes for now. We are moving to the Explain phase — your job shifts from seeing to explaining.' Assign expert groups by counting off 1–5 around the room. [Post group assignments and their structure on the board.] 'You have 6 minutes in your expert group. I want THREE things from you: what does it look like, what does it do, and how is it connected to the sukuma wiki in our phenomenon? Go.' [Circulate. At 3 minutes, prompt quietly to any group that is off-task:] 'You have 3 minutes left — have you linked your structure to the wilting yet?'	Jigsaw Expert Groups with Live Diagram Annotation: The jigsaw structure ensures every student is accountable for deep understanding of at least one structure while receiving peer-taught summaries of the others — building the full picture collaboratively. Live annotation by the teacher models how scientists build diagrams incrementally as evidence accumulates. The 15-second enforced wait time after the synthesis question is critical: it prevents fast responders from dominating and gives all learners — including EAL students — time to formulate thoughts. Connecting	Observable indicators: (1) Each expert group produces all three required statements (look, function, phenomenon link) — teacher checks during circulation. (2) In the individual 3-sentence written response, students explicitly name at least one leaf structure (e.g., stomata, guard cells, xylem) AND use the word 'water' in relation to wilting. (3) In the show-of-hands vote, convergence toward the correct explanation (stomata open → water vapour exits → turgor lost → wilting) is the target. If fewer than 60% of the class converges, the teacher re-teaches using a

Source: CK-12  Search ARES for similar readings	and one sentence connecting it to the sukuma wiki phenomenon. Groups present in sequence; the teacher annotates the projected diagram live as each group speaks. After all five presentations, the class co-constructs a summary table (structure → location → adaptation → function → connection to sukuma wiki wilting). The teacher then poses the key synthesis question: 'We now know stomata open to let CO₂ in and O₂ out. But what else moves through open stomata? And if midday sun makes stomata open wide, what does that mean for water loss?' Students write a 3-sentence response in their notebooks individually before the teacher takes a class vote using show-of-hands on the most likely explanation for midday wilting.	[After 6 minutes, call presentations:] 'Epidermis group — 60 seconds. I'll be annotating the diagram as you speak, so be precise with your location words: upper, lower, outer, inner.' After all 5 groups present, pause: 'Before we fill in the summary table, I want everyone to think silently for 15 seconds about this question' [project it on screen]: 'What else moves through an open stoma besides gases?' [15-second wait time — enforce silence explicitly: 'I am waiting for everyone to think, not just the first hands.']. Cold-call 4 students. Then say: 'Write your three-sentence explanation of midday wilting NOW — in your own words, using the structures we just named. Two minutes.' [After 2 minutes, conduct show-of-hands vote and select two contrasting responses to read aloud and compare.]	every structure back to the sukuma wiki phenomenon prevents the lesson from becoming an isolated anatomy exercise.	targeted analogy: 'Think of the stomata as a window in a nyumba — open it on a hot Nairobi afternoon and what happens to the air inside?'
Driving Question Board (DQB) Creation  VIDEO: 12 Plant Transport Xylem And Phloem, Transpiration [3D Animation] <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  HTML: Leaf Structure and Function - Advanced (Flexbook) Source: CK-12	Students return to the class Driving Question Board (established in Lesson 1). Each student receives two sticky notes. On the first (yellow), they write one NEW question that today's leaf structure observation has generated — something they now wonder about that they did not wonder about before (e.g., 'What controls whether stomata open or close?', 'Do stomata open at night?', 'Why are palisade cells packed so tightly at the top?'). On the second (green), they write one question from Lesson 1's DQB that they now feel they can partially answer based on today's evidence, and they write a brief partial answer. Students post their sticky notes on the DQB in the relevant clusters (established in	Distribute two sticky notes per student. Say: 'The DQB is our class brain — it shows what we know and what we still need to find out. You have 3 minutes: yellow note = a brand new question today's evidence gave you; green note = a question from Lesson 1 you can now partially answer. Write NOW — no talking.' [After 3 minutes:] 'Post your notes in the matching cluster on the board. If you are not sure which cluster, make a new one.' [Allow 2 minutes for posting.] 'Now — 3 minutes of silent gallery walk. Read what your classmates wrote. Put a small dot on any yellow-note question that you also wonder about.' [After gallery walk:] 'Let me read three questions that I think will drive our next three lessons.' [Read selected	Two-Colour DQB Protocol (New Questions / Partial Answers): Distinguishing between new questions (yellow) and partial answers (green) makes the growth of understanding visible — students literally see the board filling with green as the unit progresses. The dot-voting during the gallery walk democratises curiosity: students see that their peers share the same wonders, building a community of inquiry. Referencing Kenyan researchers (KARI — Kenya Agricultural and Livestock Research Organisation) grounds the questions in a real, local scientific community.	Observable indicators: (1) Every student posts at least one yellow and one green sticky note. (2) Yellow notes reference specific structures observed today (stomata, guard cells, etc.) rather than vague general questions. (3) Green notes contain a partial mechanistic answer — not just 'because of water' but 'because stomata are open and water vapour escapes.' Teacher collects and photographs the DQB at the end of each lesson to track conceptual progression across the unit.

 Search ARES for similar readings	Lesson 1). The class spends 3 minutes doing a silent gallery walk of the new additions. The teacher then reads out 2–3 of the new yellow-note questions that anticipate upcoming lesson content (stomatal opening mechanism, chlorophyll and light, photosynthesis equation) and says: 'These questions are excellent. They are exactly what we will be investigating over the next several lessons. Keep them in mind.'	questions slowly and clearly.] 'These are the questions that real plant biologists in Kenya — like researchers at KARI in Nairobi — are also asking. Hold onto them.'		
Model Building Phase  VIDEO: 12 Plant Transport Xylem And Phloem. Transpiration [3D Animation] <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria  Search ARES for similar videos  HTML: Leaf Structure and Function - Advanced (Flexbook) Source: CK-12  Search ARES for similar readings	Students retrieve their initial phenomenon model from Lesson 1 (a sketch explaining why the sukuma wiki wilts). They now REVISE this model to incorporate today's structural evidence. Using coloured pencils, they add a cross-section of a leaf showing at least: the epidermis/cuticle, palisade mesophyll, spongy mesophyll, stomata with guard cells, and a vascular bundle. They draw arrows showing: (a) light entering from above through the epidermis; (b) water moving up through the xylem; (c) water vapour exiting through the stoma. They annotate each arrow with a brief label. Below their model, students write a 'Model Revision Statement': 'In Lesson 1 I thought _____. Now I have added _____ because today's evidence showed _____. My model still cannot explain _____. The unresolved part — typically the mechanism of stomatal opening and why it is triggered by light — is circled in red as a 'placeholder for future lessons.' Models are kept in the student's personal Biology portfolio for cumulative revision throughout the unit.	Say: 'Get out your Lesson 1 model. You are about to upgrade it — like updating an app on your phone when new information becomes available. This is exactly what scientists do.' [Allow 30 seconds for students to find their Lesson 1 models.] 'You have 8 minutes. Add the leaf cross-section to your model. Add arrows for light, water, and water vapour. Write your Model Revision Statement at the bottom. If you finish early, add a second arrow showing sugars moving DOWN through the phloem — we'll explain that fully in a future lesson.' [Circulate. At 5 minutes, prompt:] 'Make sure your Model Revision Statement has all three parts — what you thought before, what you added, what you still can't explain.' [At 8 minutes:] 'Hold up your revised model — I want to see that every model now has a cross-section with arrows.' [Do a quick visual scan.] 'Excellent. Put a red circle around the part of your model you cannot yet explain. That red circle is your personal research question for this unit. We will fill it in together.' [Collect one strong model and one weak model	Cumulative Model Revision with Explicit Revision Statements: Requiring a written 'Model Revision Statement' makes the cognitive process of updating a mental model explicit and metacognitive — students articulate what changed and why, rather than simply redrawing. The 'red circle placeholder' technique normalises scientific incompleteness: it is not a failure to not yet understand stomatal mechanics; it is an honest scientific statement of the boundaries of current evidence. This directly mirrors how the sukuma wiki student's understanding is deepening — she can now explain WHAT is happening structurally, but not yet fully WHY stomata open in response to light.	Observable indicators: (1) Revised model includes a recognisable leaf cross-section with at least 3 named layers. (2) At least two directional arrows are present with labels (water up, water vapour out, or light in). (3) Model Revision Statement contains all three required components — prior belief, new addition, unresolved question. (4) The 'red circle' is present and the unresolved question is mechanistic (e.g., 'I still can't explain what makes the guard cells open the stoma') rather than trivial. Teacher uses the two anonymously collected models at the start of Lesson 3 as a 'Model Clinic' to sharpen the class's modelling skills.

		— anonymously — to display next lesson as 'Model A' and 'Model B' for the class to compare and improve.]		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the Observe phase, how many pairs successfully located stomata on the lower epidermis under the hand lens or microscope? If fewer than half succeeded, consider preparing pre-mounted slides or a printed photomicrograph as a fallback for Lesson 3. (2) In the Explain phase, did expert groups make the three-part connection (structure → function → phenomenon) or did they stop at function? If connections to the wilting phenomenon were weak, open Lesson 3 with a targeted question: 'Yesterday you described what the stomata do — today I want us to explain WHY that matters to our sukuma wiki.' (3) Were yellow DQB sticky notes genuinely mechanistic (e.g., 'What makes guard cells swell?') or superficial (e.g., 'Why are plants green?')? Mechanistic questions signal readiness for Lesson 3's deeper dive into chloroplasts and pigments. (4) Review 5–6 revised models at random: are arrows directional and labelled? Are Model Revision Statements specific? Students who write 'I still don't understand photosynthesis' need more scaffolding in their revision statements — prompt them to name a specific structure they observed that they cannot yet fully explain. (5) Was the 15-second wait time enforced consistently? Note which students have not yet voluntarily contributed to whole-class discussion and cold-call them specifically in Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students examined the internal cross-section of a dicot leaf (mango or bean) using a hand lens and microscope, identifying the upper epidermis, cuticle, palisade mesophyll, spongy mesophyll, air spaces, lower epidermis, stomata with guard cells, and vascular bundles (xylem and phloem).
What did I learn?	Each leaf layer is structurally adapted to a specific function: the transparent cuticle/epidermis admits light while reducing water loss; the densely packed palisade mesophyll maximises chloroplast exposure to light; the spongy mesophyll provides air spaces for CO ₂ and O ₂ diffusion; stomata regulate gas exchange and water vapour loss; xylem transports water and minerals upward while phloem transports sugars downward.
How does this explain the phenomenon?	The sukuma wiki wilts at midday even in moist soil because strong light triggers stomata to open wide for gas exchange (CO ₂ in, O ₂ out), simultaneously allowing rapid water vapour loss (transpiration) through the lower epidermis. When the rate of water vapour loss through open stomata exceeds the rate at which xylem can deliver water from the roots, leaf cells lose turgor pressure and the plant wilts. At night, stomata close, transpiration stops, xylem catches up, turgor is restored, and the plant recovers — explaining the daily wilting-and-recovery cycle.

LESSON 3 (40 minutes): Raw Materials for Photosynthesis: What Does a Plant Actually Need?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners identify the four raw materials required for photosynthesis, write the word and chemical equations, and use those raw materials to begin explaining why the sukuma wiki wilts at midday and turns pale in shade.
Knowledge	Learners will be able to: (1) state the four raw materials for photosynthesis — carbon dioxide, water, light energy, and chlorophyll; (2) write the word equation for photosynthesis; (3) write the balanced chemical equation for photosynthesis; (4) identify which raw materials are implicated in midday wilting and which are implicated in pallor under shade.
Skills	Learners will practise structured reading for information extraction, guided scientific discussion, equation writing, and evidence-based reasoning to link raw materials to observable symptoms in the anchoring phenomenon.
Attitudes	Learners develop curiosity about how everyday observations — such as a wilting kitchen-garden plant — can be explained through chemistry and biology, and appreciate the relevance of plant science to Kenyan food security and agriculture.
Key Inquiry Question	How do plants make their own food? (KICD Key Inquiry Question 1)
Purpose in Storyline	Lesson 2 revealed the internal architecture that enables photosynthesis. Lesson 3 now identifies WHAT the plant must gather to run photosynthesis, providing the conceptual bridge between structure and process. This makes the wilting and pallor symptoms causally explicable — wilting relates to water and high light-driven transpiration, pallor relates to reduced light and therefore reduced chlorophyll synthesis — setting up the starch-test evidence in Lesson 4.
Safety Notes	No hazardous chemicals are used in this lesson. If printed text resources are distributed, ensure they are legible. Learners with visual impairments should be seated near the board. No special safety precautions are required beyond standard classroom conduct.

B. LESSON OVERVIEW

The lesson opens by revisiting the sukuma wiki phenomenon and asking learners to predict what inputs the plant needs before any structured reading occurs. Learners then work in pairs through a two-page structured reading resource (provided as a printed handout or written on the board in resource-limited settings) that presents evidence for each raw material — CO₂ entry via stomata, water uptake via roots and xylem, light absorption by chlorophyll, and the role of chlorophyll as a pigment. Guided whole-class discussion consolidates understanding. Learners write the word equation and chemical equation in their notebooks, then apply their new knowledge back to the phenomenon by categorising which raw materials explain wilting and which explain pallor on the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Photosynthesis in organisms Source: Khan	Learners open their notebooks to their Lesson 1 predictions. The teacher displays or reads aloud two sentences from the original phenomenon description: (1) The	Display or write the two phenomenon sentences on the board. Distribute notebooks prompt or write it on the board: List at least two things you think the	Predict-then-read structure activates prior knowledge from Lesson 2 (leaf anatomy) and creates cognitive hooks for the reading. Recording predictions	Teacher scans the board columns for misconceptions — for example, learners who list soil nutrients as the only input, or who omit light. These gaps are noted mentally to

Academy (English - US curriculum) Search ARES for similar videos  HTML: Light Reactions of Photosynthesis (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	sukuma wiki leaves wilt at midday even though the soil is moist. (2) In shade the wilting stops but the leaves turn yellowish-green and growth slows. Each learner writes a response to the prompt: List at least two things you think the plant needs in order to stay green and keep growing. Give a reason for each thing you list. Learners share with a partner before two or three pairs volunteer answers to the class. Responses are recorded in two columns on the board under the headings NEEDED TO STAY GREEN and NEEDED TO GROW. No answers are confirmed or rejected yet — they are treated as hypotheses to be tested against the reading.	plant needs to stay green and keep growing — give a reason for each. Say: Think back to the leaf structure we studied last lesson. You saw where water comes in, where light enters, and where gases move. Use that knowledge to predict what raw materials photosynthesis might need. Allow 2 minutes of individual silent writing. Then say: Compare your list with your partner — do you agree? You have 1 minute. [WAIT TIME: 1 minute pair discussion.] Cold-call two pairs. Record their suggestions in the two columns without evaluating. Say: We are going to read some evidence now that will help us decide which of these predictions are correct and which need to change.	publicly makes thinking visible and creates accountability for revision during sense-making.	probe during the Explain phase. Teacher circulates during pair discussion and listens for reasoning quality.
Observe Phase  VIDEO: Photosynthesis in organisms Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Light Reactions of Photosynthesis Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners receive the structured reading handout titled Evidence for the Raw Materials of Photosynthesis (or follow along as the teacher reads from the board in low-resource settings). The handout contains four short evidence paragraphs, one for each raw material, each linked to a real Kenyan context: (A) Carbon dioxide — scientists at the University of Nairobi have measured CO₂ concentrations inside and outside leaf chambers; CO₂ enters through stomata as shown in Lesson 2. (B) Water — water travels from Nairobi City Water and Sewerage Company boreholes through soil into plant roots, up the xylem, into mesophyll cells; without water the stroma cannot carry out reactions. (C) Light energy — solar radiation data from the Kenya Meteorological Department shows peak irradiance in Nairobi between	Distribute handouts or direct attention to the board. Say: Read each paragraph carefully. As you read, circle every raw material you find, underline where or how the plant gets it, and put a question mark next to anything confusing. You have 6 minutes to read and annotate silently. [WAIT TIME: 6 minutes silent reading.] After 6 minutes say: Turn to your partner — share one thing you circled and one thing you put a question mark next to. [WAIT TIME: 2 minutes pair share.] Then ask the class: What question marks came up? Collect two or three genuine confusion points and write them on the board labelled QUESTIONS FROM THE READING. Say: We will address every one of these during our discussion right now.	Structured annotation (circle, underline, question mark) is a disciplinary literacy strategy that trains learners to read scientific text actively rather than passively. Collecting question marks publicly validates confusion as a scientific starting point and models how scientists track unknowns.	Teacher circulates during silent reading and checks annotations. Are learners circling all four raw materials or only the obvious ones? Are question marks clustered around the chemical equation or around a specific material? This informs the depth of explanation needed in the Explain phase.

	11:00 and 14:00, coinciding exactly with the wilting window observed by the student. (D) Chlorophyll — the green pigment in chloroplasts absorbs red and blue light; plants in shade receive less light so less chlorophyll is stimulated and, over time, less chlorophyll is synthesised, causing pallor. Learners read silently and annotate using the code: circle each raw material name, underline the sentence that tells you where or how the plant gets it, and place a question mark next to anything they do not understand.			
Explain Phase  VIDEO: Photosynthesis <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Photosynthesis and Energy for Cells Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	The teacher leads a 12-minute guided discussion that moves through three steps: (1) Confirming the four raw materials — CO₂, H₂O, light energy, chlorophyll — with learners adding or correcting their prediction lists. (2) Building the word equation together: learners suggest inputs and outputs; the teacher scribes on the board: carbon dioxide plus water plus light energy in the presence of chlorophyll yields glucose plus oxygen. Learners copy this into their notebooks. (3) Building the chemical equation: the teacher writes the unbalanced equation on the board and invites learners to count atoms together — 6CO₂ plus 6H₂O yields C₆H₁₂O₆ plus 6O₂ with light energy and chlorophyll shown above the arrow. Learners copy the balanced equation and label each term. The teacher then poses the application question: Looking at the four raw materials, which one or ones do you think are most directly responsible for the wilting at midday? Which are responsible for the pallor in shade? Learners	Return to the prediction columns on the board. Say: Let us check your predictions. Which of these items appear in what you read? Cross out anything the evidence does not support and add anything you missed. Give learners 1 minute to update their notebooks. Then say: I am going to write a sentence on the board and I need you to help me complete it. The raw materials for photosynthesis are — pause — who can give me the first one? Build the word equation collaboratively, scribing exactly what learners say and correcting gently. Say: Now I want to write this as a chemical equation. Has anyone seen chemical formulas before? Do not worry if not — we will build it together by counting atoms. Work through the balancing step by step, narrating your thinking aloud: I need 6 carbons on the left so I use 6CO₂. After the equation is complete, say: Now the big question — and I want you to think carefully before you answer. [WAIT TIME: pause 5 seconds of silence.] Of these four raw	Collaborative equation building (rather than teacher telling) positions learners as co-constructors of knowledge. The application question creates a bridge between abstract chemistry and the lived phenomenon, using the leaf structure knowledge from Lesson 2 as a reasoning scaffold. The explicit connection — light drives both photosynthesis and transpiration — is a high-leverage insight that advances the storyline.	Teacher listens to pair discussions and cold-call responses to assess whether learners can reason causally (not just recall). Probe questions: Why does the plant not simply close its stomata during the day? What would happen to CO₂ supply if stomata closed? These questions reveal depth of understanding and surface the stomatal dilemma that will resurface in Lesson 5.

	discuss in pairs for 90 seconds before sharing answers. The teacher guides the class to the following reasoning: wilting is linked to water (high transpiration rate at peak solar irradiance exceeds water supply to leaves — even though soil is moist, the rate of loss outpaces uptake) and to light intensity driving transpiration; pallor and slow growth are linked to reduced light energy reaching chlorophyll and therefore reduced glucose synthesis.	materials, which one is most directly connected to the wilting? Which to the pallor? Discuss with your partner. [WAIT TIME: 90 seconds.] Cold-call two pairs. Probe further: How does more sunlight lead to more water loss — think about the stomata you saw in Lesson 2. Guide learners to connect stomatal opening at high light intensity to increased transpiration. Say: This is powerful — the same light that drives photosynthesis also drives the transpiration that causes wilting. That is a real scientific tension.		
Driving Question Board (DQB) Creation  VIDEO: Photosynthesis in ecosystems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: The light-dependent reactions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner receives two sticky notes (or tears a small piece of paper into two). On the first note they write one question that has been answered today about why sukuma wiki wilts or turns pale. On the second note they write one new question the lesson has raised for them. Learners post their notes in the appropriate sections of the class DQB — an ANSWERED section and a NEW QUESTIONS section. The teacher reads three to five new questions aloud and tags them with lesson numbers from the storyline to show the class when those questions will be addressed. The class identifies the two most important unanswered questions by show of hands and circles them on the DQB.	Distribute two sticky notes per learner. Say: On the first note write one question that you feel you can now answer because of what we learned today — start it with the words: I can now explain why... On the second note write one brand-new question this lesson has raised for you — something you are now curious about that you were not thinking about before class. Give 2 minutes for writing. [WAIT TIME: 2 minutes.] Direct learners to post their notes. Read three new questions aloud. Say: These are excellent scientific questions. Let me show you where in our unit each of these will be answered. Point to the storyline summary posted on the wall. Tag the oxygen question to Lesson 5, the glucose question to Lesson 6, the mineral question to Lesson 9. Say: Every question you have is part of the science of this unit. We will get to all of them.	The DQB move makes the storyline progression visible and emotionally rewarding — learners see their confusion transforming into answered questions across lessons. Labelling new questions with future lesson numbers builds anticipation and coherent narrative across the unit.	The ANSWERED sticky notes are the primary formative evidence. Teacher reads them after class to check whether learners are correctly attributing wilting to water loss and light-driven transpiration and pallor to reduced light and chlorophyll activity. Misconceptions on these notes inform the opening review of Lesson 4.
Model Building Phase  VIDEO:	In the final 5 minutes learners open to a fresh notebook page and draw a quick rough diagram — not a neat illustration but a thinking	Say: I want you to build the first version of a model you will keep adding to all the way to Lesson 10. Open a fresh page — label it MY	The living model strategy (a diagram that is deliberately incomplete and revised across lessons) embeds the	Teacher does a quick 30-second gallery walk during the drawing phase, checking that learners have placed chlorophyll inside the

Comparing animal and plant cells <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Light Reactions of Photosynthesis (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	tool — showing the sukuma wiki plant with four arrows pointing inward, each labelled with one raw material and where it enters or comes from: CO₂ from the air through stomata, H₂O from the soil through roots and xylem, light energy from the sun onto the leaf surface, and chlorophyll inside the chloroplasts of the palisade mesophyll. They add two output arrows labelled glucose and oxygen. Below the diagram they write one sentence: This diagram shows that without enough water or light reaching the leaf the plant cannot complete photosynthesis. Learners are told this rough model will be expanded in every subsequent lesson until Lesson 10 when it becomes their final annotated explanation of the whole phenomenon.	PHOTOSYNTHESIS MODEL — VERSION 1. Draw a simple outline of the sukuma wiki plant. Add four arrows for the raw materials coming IN and two arrows for the products going OUT. Label everything. You have 4 minutes. [WAIT TIME: 4 minutes.] As learners draw, circulate and check that all four inputs are present and correctly sourced. After 4 minutes say: Write one sentence below your diagram beginning with: This diagram shows that... Cold-call one learner to read their sentence. Affirm and correct gently. Say: Next lesson we will use iodine solution and ethanol to test a real leaf and find out whether photosynthesis actually produced starch — and your model will get its first piece of experimental evidence.	metacognitive habit of treating scientific knowledge as provisional and cumulative. It also creates a personal artefact that spans the entire unit, making the storyline tangible for each learner.	chloroplast (not outside the plant) and that xylem is mentioned as the water pathway — two concepts from Lesson 2 that should now be integrated. Any learner who omits chlorophyll or places it incorrectly receives a quiet one-to-one prompt before the lesson ends.
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D. TEACHER REFLECTION

After the lesson ask yourself: (1) Could learners reason causally about which raw material causes wilting versus pallor, or did they simply recall the four items as a list? If causal reasoning was weak, begin Lesson 4 with a quick Think-Pair-Share on the question: Why does MORE sunlight cause LESS water in the leaf? (2) Were the chemical equation steps scaffolded enough, or did equation-writing create anxiety that distracted from the conceptual goals? If so, consider providing a partially completed equation frame in Lesson 4 activities. (3) Did the DQB new-question notes reveal any persistent misconceptions — for example, that chlorophyll is a raw material consumed by the reaction rather than a catalyst-like pigment? If so, address this at the opening of Lesson 5 when chlorophyll is discussed in the context of light absorption. (4) Were Kenyan contexts (Kenya Meteorological Department irradiance data, University of Nairobi research) engaging and credible to learners, or did they feel forced? Adjust the framing in future lessons based on learner response.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We read evidence showing that carbon dioxide enters the leaf through stomata, water travels from the soil up the xylem to mesophyll cells, light energy from the sun peaks in Nairobi between 11:00 and 14:00 coinciding with wilting, and chlorophyll in the palisade mesophyll absorbs that light.
What did I learn?	Photosynthesis requires four raw materials: carbon dioxide, water, light energy, and chlorophyll. The word equation is: carbon dioxide plus water in the presence of light energy and chlorophyll yields glucose plus oxygen. The balanced chemical equation is: 6CO₂ plus 6H₂O yields C₆H₁₂O₆ plus 6O₂. Chlorophyll is not consumed in the reaction but is essential as a light-absorbing pigment.

How does this explain the phenomenon?

The sukuma wiki wilts at midday because peak solar irradiance causes stomata to open wide for CO₂ uptake, which simultaneously increases transpiration so water leaves the leaf faster than the xylem can replace it — even when soil is moist. The pallor and slow growth in shade are explained by reduced light energy reaching chlorophyll, which reduces the rate of photosynthesis and therefore reduces glucose production, slowing growth and reducing the stimulus for chlorophyll synthesis.

LESSON 4 (40 minutes): Starch Test Practical: Finding Evidence That Chlorophyll and Light Are Both Needed for Photosynthesis

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners carry out the standard starch test on a variegated or partially shaded leaf to generate direct observational evidence that both chlorophyll and light are required for photosynthesis to produce starch.
Knowledge	Learners will be able to: (1) state that starch is a product of photosynthesis stored in leaves; (2) describe the steps of the starch test using hot ethanol and iodine solution; (3) explain that blue-black colouration in the iodine test indicates the presence of starch; (4) conclude that only regions of a leaf that are green AND were illuminated produce starch, providing evidence that both chlorophyll and light are necessary for photosynthesis.
Skills	Safe laboratory technique including handling hot water baths and ethanol; careful observation and recording of colour changes; scientific inference from experimental results; annotating diagrams of leaf results.
Attitudes	Curiosity and open-mindedness when results challenge prior assumptions; honesty in recording actual observations rather than expected ones; appreciation of practical evidence as the foundation of scientific knowledge.
Key Inquiry Question	How do plants make their own food? (Key Inquiry Question 1) — this lesson provides physical evidence from inside the leaf itself that answers part of this question.
Purpose in Storyline	Lessons 1 to 3 built the conceptual framework: the sukuma wiki wilts because of water and light, and requires CO ₂ , H ₂ O, light, and chlorophyll to photosynthesise. This lesson shifts from reading and discussion to hands-on evidence generation. Learners see with their own eyes that removing chlorophyll (white sections of a variegated leaf) or removing light (covered section) both prevent starch production. This is the first concrete, self-generated evidence in the sequence and directly justifies why the pale, shaded sukuma wiki produces less food. It also sets up Lessons 5 and 6 which explain the biochemical mechanism behind these observations.
Safety Notes	SAFETY: Ethanol is highly flammable — NO naked flames when ethanol is in use. Use an electric hot plate or a pre-boiled hot water bath maintained away from the burner. Learners must wear eye protection when handling iodine solution and hot water. Hot ethanol tubes must be handled with tongs or a cloth. Iodine solution stains skin and clothing — warn learners. If a variegated plant (e.g., Coleus or variegated Sansevieria) is not available, a bean or kale leaf partially covered with black paper or aluminium foil for 48 hours before the lesson is an equally valid and locally accessible alternative.

B. LESSON OVERVIEW

Learners begin by predicting which parts of a variegated or partially shaded leaf they expect to contain starch and why, linking back to the raw materials identified in Lesson 3. They then carry out the classic starch test — killing the leaf in boiling water, decolourising in hot ethanol, washing, and applying iodine solution — and record colour changes on a diagram. In the Explain phase they match their results to the prediction, construct a conclusion linking chlorophyll and light to starch production, and update the Driving Question Board with new evidence. A brief model-building task asks learners to annotate a simple leaf cross-section diagram to show where photosynthesis is and is not occurring based on their results.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Photosynthesis in ecosystems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: The light-dependent reactions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners are shown either (a) a real variegated leaf (e.g., from a Coleus plant available in many Nairobi school gardens or a variegated Sansevieria from a household) placed on a white sheet of paper, OR (b) a diagram of a leaf showing a white/yellow region, a green region, and a green region that was covered with black paper for 48 hours. Working individually for 2 minutes and then sharing with a partner, each learner draws and colours the leaf in their notebook, labelling the three zones, and writes a prediction answering: Zone 1 — white/no-chlorophyll region: will starch be present? Zone 2 — green uncovered region: will starch be present? Zone 3 — green covered region: will starch be present? Learners must justify each prediction using the word equation from Lesson 3, explicitly naming which raw material is absent in each zone.	Display the leaf or diagram on the board. Say: 'Before we do any chemistry, I want you to think like a scientist. Look at this leaf carefully. It has three different zones. Using only what we learned last lesson about the raw materials for photosynthesis, predict in your notebook which parts you expect to contain starch after our test — and give a reason for each one.' Allow 2 minutes of silent individual writing. WAIT TIME: Do not speak or prompt during these 2 minutes. Then say: 'Share your prediction with the person next to you. Do you agree? If not, whose reasoning is stronger?' After 1 minute of pair talk, cold-call two or three pairs to share contrasting predictions. Record the class predictions in two columns on the board — STARCH PRESENT and NO STARCH — against each zone. Say: 'We are going to keep these predictions visible on the board. At the end of the practical, we will come back and see who the evidence supports.'	Prediction-justification routine forces learners to apply prior knowledge (raw materials from Lesson 3) to a new situation. Recording predictions publicly on the board creates accountability and cognitive dissonance when results differ from expectations, which drives deeper explanation in the Explain phase.	Teacher circulates during individual writing and reads predictions over shoulders. Listen for: (1) correct reasoning — white zone lacks chlorophyll so no starch; covered zone lacks light so no starch; green uncovered zone has both so starch present; (2) misconceptions — some learners may predict that the covered green zone still makes starch because it is still green (confusing chlorophyll presence with chlorophyll function). Note these learners for targeted questioning during the Explain phase.
Observe Phase  VIDEO: Photosynthesis in organisms <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Light Reactions of Photosynthesis	Groups of 3 to 4 carry out the starch test following a step-by-step method card provided by the teacher. Steps: (1) Place the leaf in boiling water for 2 minutes to kill cells and stop enzyme activity — observe that the leaf becomes limp and darker. (2) Transfer the leaf using tongs into a tube of ethanol placed in a hot water bath (ethanol must NOT be heated directly over a flame) for 5 to 7 minutes until the leaf is decolourised and the ethanol turns green — observe the colour of both the leaf and the ethanol. (3) Remove the leaf, rinse	Before the practical begins, demonstrate Step 1 and Step 2 with one leaf while the class watches, narrating the safety points: 'Notice I am keeping the ethanol tube in the hot water bath — never over a flame. I am using tongs, not my fingers, to handle the hot tube. Everyone, check that your burner or hot plate is switched off before you pick up any ethanol.' Circulate constantly during the practical. At the decolourisation step, prompt: 'What colour is the ethanol turning? Why do you think that is	Real-time observation of colour change in a self-conducted experiment creates high cognitive engagement. The instruction to record even surprising results builds scientific honesty. Comparing the physical leaf to the board predictions creates the conceptual tension that drives the Explain phase.	Check results tables as groups complete them. Correct errors immediately — for example, if a group records the covered green zone as blue-black, ask them to check whether that section was truly covered (it may not have been, which is itself a useful discussion point about experimental control). Listen for spontaneous explanations among learners — these reveal depth of understanding. Ask any group that finishes early: 'If we had used a completely yellow leaf with no chlorophyll anywhere, what result

(Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	gently in warm water to soften it, and spread flat on a white tile or petri dish. (4) Flood the leaf with iodine solution for 1 to 2 minutes. (5) Observe and record the colour of each zone immediately. Learners record results by: (a) writing the colour of each zone in a results table (Zone, Colour after iodine, Starch present YES or NO); (b) drawing and colouring an accurate diagram of the leaf after iodine treatment in their notebooks.	happening?' WAIT TIME: pause 10 seconds after this question before accepting answers. At the iodine step, instruct learners to apply iodine and watch in real time: 'Do not look away — the colour change happens quickly. Record exactly what you see, even if it surprises you.' After results are recorded, ask: 'Compare your leaf diagram to the predictions still on the board. What do you notice?' Allow 1 minute of silent comparison before moving to the Explain phase.		would you predict? What about a completely green leaf that had been in the dark for 48 hours?
Explain Phase  VIDEO: 02 Photosynthesis Light And Dark Reactions <i>Related topic: not an exact match for this lesson.</i> Source: Seavuria Search ARES for similar videos  EXERCISE: Photosynthesis <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Whole-class discussion and individual written conclusion. The teacher guides learners to construct a three-part evidence-based conclusion by working through the results zone by zone. Learners then write the conclusion in their notebooks in the following structured format: OBSERVATION (what colour appeared), INFERENCE (what this tells us about starch), CONCLUSION (what this tells us about the requirement for photosynthesis). After the structured conclusion, learners revisit the sukuma wiki phenomenon: they write one sentence connecting their practical result to why the shaded sukuma wiki turned pale and grew slowly — linking lack of light to reduced starch and glucose production, which slows growth. Finally, a guided question asks: 'Our result shows that BOTH chlorophyll and light are needed. From Lesson 3, we know CO₂ and water are also needed. How would you design an experiment to test whether CO₂ is essential?' Learners briefly discuss in pairs.	Project or draw a three-column table on the board: Zone, Colour after iodine, Conclusion. Work through the table as a class. For the white zone, say: 'Everyone who got no colour change here — what does that tell us? What was missing in this zone?' WAIT TIME: 10 seconds. Accept answers. Write the class consensus. Then say: 'So removing the green pigment alone was enough to stop starch production. What does that prove about chlorophyll?' For the covered green zone, say: 'This zone had chlorophyll — it was still green — but it was kept in the dark. What happened?' WAIT TIME: 10 seconds. Guide learners to state: light is required even when chlorophyll is present. Say: 'This is important. Chlorophyll alone is not enough. Light alone is not enough. Both must be present at the same time.' Write this on the board as a key statement. Then connect to the phenomenon: 'Remember our sukuma wiki under the mango tree — it had chlorophyll, but what was reduced?' Elicit: light intensity. Say: 'And our results just proved	Structured conclusion writing (Observation - Inference - Conclusion) scaffolds scientific reasoning and prevents learners from writing vague statements. Connecting the result to the phenomenon at the end of the Explain phase maintains the storyline thread. The extension question about CO₂ foreshadows Lesson 7 (the investigation) and keeps inquiry open.	Collect written conclusions at the end of the lesson or photograph them for review. Look for: (1) correct use of evidence — does the learner cite the actual colour change observed, not just state what they expected? (2) correct causal language — does the learner say chlorophyll is REQUIRED rather than just present? (3) connection to the phenomenon — can the learner transfer the lab result to the sukuma wiki context? Common error to watch for: learners writing that the white zone turned blue-black because it had no starch — they must explain WHY it had no starch (no chlorophyll, so no photosynthesis).

		that reduced light means reduced starch production. That is why the shaded plant grew slowly.' Close by acknowledging the DQB: 'We now have laboratory evidence, not just a hypothesis, for part of our driving question. Let us update the board.'		
Driving Question Board (DQB) Creation  VIDEO: Breaking down photosynthesis stages <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: The light-dependent reactions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each group receives two sticky notes (different colours: one for EVIDENCE WE NOW HAVE, one for NEW QUESTIONS RAISED). Groups write: on the first note, one piece of evidence from today's practical that helps answer the driving question; on the second note, one new question the practical has raised (for example: If chlorophyll absorbs light, which colours of light work best? What exactly does chlorophyll do with light energy?). Groups post their notes on the class DQB under the appropriate columns. The teacher reads selected notes aloud and the class votes on which new question they most want to answer, flagging it for Lesson 5.	Point to the DQB and say: 'In Lesson 1 we put up questions about why sukuma wiki wilts and why shade causes pallor. In Lesson 3 we started answering some of those. Today we have real experimental evidence. Take 2 minutes — one evidence note, one new question note.' WAIT TIME: allow genuine silent writing for 2 minutes. As groups post notes, read two or three aloud. Say: 'Look at the new questions people are asking — they want to know what chlorophyll actually DOES with light. That is exactly what we will investigate in Lesson 5. Science works this way: answering one question always opens new ones.' Leave all notes visible for the remainder of the unit.	The DQB routine makes the epistemic progress of the class visible — learners see questions being answered and new ones generated, modelling authentic scientific practice. The voting on the next question gives learners agency in the inquiry sequence.	Read all posted evidence notes. Check that evidence notes cite the starch test result specifically, not just general statements. If any note says only 'plants need light' without referencing the experimental observation, ask that group to revise: 'What did you actually see in the iodine test that tells you that?' This ensures learners are reasoning from evidence rather than recalling prior knowledge.
Model Building Phase  VIDEO: Photosynthesis: Overview of the light-dependent reactions <i>Related topic: not an exact match for this lesson.</i>	Individual task (5 minutes). Learners receive or draw a simple outline of a leaf cross-section with three zones marked (matching the three zones of the variegated or covered leaf). Using coloured pencils, learners annotate: (1) which zones contain chlorophyll; (2) which zones received light; (3) which zones produced starch (shade with blue-black cross-	Say: 'I want you to build a small model in your notebook that captures everything we just proved. Draw or trace the leaf outline I have given you. Use colour to show your results, and then explain each zone in writing using the word equation. This diagram will be the starting point for the bigger model we build at the end of the unit.' Circulate and	The annotated cross-section diagram serves as a cumulative model artefact — learners will add the light-dependent and light-independent reactions to it in Lessons 5 and 6, and it will form the basis of the final synthesis model in Lesson 10. Framing it as a growing model sustains motivation across the unit.	Check that annotations correctly assign the absence of starch to the correct missing variable in each zone. A learner who labels the covered green zone as lacking starch because of no chlorophyll has a misconception that must be corrected before Lesson 5. Exit check: ask two or three learners to read their label for Zone 3 (covered green) aloud — the label

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Light Reactions of Photosynthesis (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	hatching); (4) a brief written label for each zone explaining why starch is or is not present, using the word equation from Lesson 3 as a reference. This annotated diagram becomes part of their running model that will be extended in Lessons 5, 6, and 10.	prompt: 'Show me which zone proves that chlorophyll alone is not enough. Now show me which zone proves that light alone is not enough.' WAIT TIME: after each prompt, pause 8 seconds before offering further guidance. Close the lesson by returning to the phenomenon: 'Next lesson we will ask: what is chlorophyll actually doing with that light — and that will finally start to explain the oxygen bubbles and the energy the sukuma wiki needs to grow.'	must reference light as the missing factor, not chlorophyll.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did all groups achieve clear decolourisation? If not, was the ethanol bath hot enough and was sufficient time allowed? Incomplete decolourisation makes iodine results ambiguous — plan to pre-prepare decolourised leaves as backup. (2) Did learners distinguish between chlorophyll being PRESENT and chlorophyll being FUNCTIONAL — this is a subtle but important conceptual point that underpins Lesson 5. (3) Were the written conclusions evidence-based or did learners just repeat what they predicted? If learners struggled to write conclusions from evidence, build in a sentence-starter scaffold next time: 'The result showed that the [zone] turned [colour], which means starch was [present or absent], which proves that [raw material] is required for photosynthesis because...' (4) Which DQB questions about what chlorophyll does with light were most frequently asked? Use these to personalise the opening of Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	When iodine solution was applied to a decolourised variegated or partially covered leaf: the white region (no chlorophyll) remained brown-orange; the covered green region (chlorophyll present but no light) remained brown-orange; the green uncovered region (chlorophyll present and light available) turned blue-black, indicating starch was present only in that zone.
What did I learn?	Both chlorophyll AND light must be present simultaneously for photosynthesis to occur and for starch to be produced in a leaf. Removing either one — even if the other is present — is sufficient to prevent photosynthesis. This is direct experimental evidence, not just a stated fact.
How does this explain the phenomenon?	The shaded sukuma wiki in the phenomenon turns pale and grows slowly because, even though it retains chlorophyll, reduced light intensity under the mango tree limits photosynthesis and therefore reduces the glucose and starch available for growth — consistent with the starch test result showing that the covered green zone (chlorophyll present, light absent) produces no starch.

LESSON 5 (80 minutes): Light-Dependent Reactions: Where the Sun's Energy Enters the Plant

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To trace the light-dependent reactions of photosynthesis — including light absorption by chlorophyll, photolysis of water, production of ATP and NADPH, and release of oxygen — and connect these molecular events to the wilting, recovery, and pallor observed in the sukuma wiki phenomenon.
Knowledge	By the end of this lesson, learners should be able to: (a) describe the structure of the thylakoid membrane as the site of light-dependent reactions; (b) explain the absorption of light energy by chlorophyll pigments; (c) describe the photolysis of water and identify its products; (d) explain the production of ATP (via photophosphorylation) and NADPH; (e) state that oxygen released during photolysis is the source of atmospheric O ₂ .
Skills	By the end of this lesson, learners should be able to: (a) annotate a diagram of the thylakoid membrane to show the sequence of light-dependent reactions; (b) construct a cause-and-effect chain linking light intensity → chlorophyll excitation → photolysis → ATP/NADPH production → oxygen release; (c) interpret oxygen-bubble data from Lesson 4 (Elodea experiment) using the light-dependent reaction framework.
Attitudes	Learners appreciate that microscopic molecular events inside a single leaf cell explain observable, everyday phenomena such as the wilting and recovery of sukuma wiki, and develop curiosity about how living systems transform energy.
Key Inquiry Question	How do plants make their own food? What factors affect the rate at which plants make food?
Purpose in Storyline	Lesson 5 is the first EXPLAIN lesson in the sequence. Learners have spent Lessons 1–4 gathering evidence: they predicted why sukuma wiki wilts, observed the phenomenon, tested variables that affect photosynthesis rate (light, CO ₂ , temperature), and counted oxygen bubbles produced by Elodea under different light intensities. Now they open the 'black box' of the light reactions to understand the molecular mechanism that converts light energy into chemical energy — explaining WHY more light produces more oxygen bubbles and WHY the shaded sukuma wiki becomes pale and grows slowly.
Safety Notes	No hazardous chemicals in this lesson. If printed diagrams are used, ensure scissors are handled safely. If a torch or lamp is used for demonstration, do not direct the beam into learners' eyes. Remind learners to handle printed materials and models respectfully as shared resources.

B. LESSON OVERVIEW

In this EXPLAIN lesson, learners transition from evidence-gatherer to sense-maker. The teacher anchors the lesson in the oxygen-bubble data from Lesson 4 and the unresolved puzzle of the sukuma wiki phenomenon. Using a guided annotated diagram of the thylakoid membrane, learners trace the sequential steps of the light-dependent reactions: (1) absorption of light by chlorophyll in Photosystems II and I; (2) excitation and transfer of electrons along the electron transport chain; (3) photolysis of water ($2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$) as the electron source; (4) chemiosmotic synthesis of ATP (photophosphorylation); and (5) reduction of NADP⁺ to NADPH. Learners use a cause-and-effect chain graphic organiser to connect each step, then re-examine their Elodea bubble counts to explain the relationship between light intensity and reaction rate. The lesson closes with a DQB update, linking the molecular events to both the supporting phenomenon (oxygen bubbles) and the anchoring phenomenon (midday wilting, shade-induced pallor, overnight recovery of sukuma wiki).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Photosynthesis: Overview of the light-dependent reactions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: The light-dependent reactions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners individually review their Lesson 4 data table (oxygen bubble counts at different light intensities from the Elodea experiment) for 2 minutes. They then write a 2-sentence response in their science notebooks to the prompt: 'You observed that more bubbles formed when the lamp was closer to the Elodea. What do you think is happening INSIDE the leaf cell that produces those bubbles — and where exactly do you think it happens?' Learners share their predictions with a partner (Think-Pair-Share) before 3–4 pairs share with the class. Predictions are posted on the class whiteboard under the heading 'Our Current Best Ideas' to be revisited at the end of the lesson.	Begin by projecting the class Elodea bubble-count data table compiled from Lesson 4. Say: 'Last lesson we showed that the more light an aquatic plant receives, the more oxygen it releases. Today we are going to open the black box — we are going to find out exactly what is happening inside the chloroplast to make that oxygen.' Distribute Lesson 4 data notebooks or display the class data. Give the written prompt verbally and write it on the board. Allow 2 minutes of SILENT individual writing — use a visible timer. After 2 minutes, say: 'Now share your prediction with your partner. You have 90 seconds.' After partner talk, cold-call 4 pairs (vary by gender and seating zone): 'Amina, what did you and your partner predict?' After each response, ask a probing question: 'What evidence from the Elodea experiment makes you think that?' Do NOT correct predictions — record all ideas on the board. Say: 'Hold these ideas. By the end of class you will be able to confirm, revise, or reject each one with evidence.'	Think-Pair-Share with written anchor: Requiring learners to write before speaking ensures all learners commit to a prediction, not just confident volunteers. Posting predictions publicly creates accountability and sets up a cognitive 'return' at the end — learners must revisit and revise their own thinking, which is the core of sensemaking.	Teacher circulates during silent writing and scans notebooks. Indicators of readiness: (a) learner references specific data (e.g., 'at 10 cm distance we got 45 bubbles per minute'); (b) learner uses mechanistic language ('something breaks down water' or 'chlorophyll captures light'). Red flag: learner writes only 'photosynthesis happens' with no mechanism — teacher makes a note to pair this learner with a stronger explainer during the diagram activity.
Observe Phase  VIDEO: Photosynthesis: Overview of the light-dependent reactions - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES	Learners receive a printed half-page diagram showing a cross-section of a thylakoid membrane with labelled but unexplained components: Photosystem II (PSII), Photosystem I (PSI), the electron transport chain (ETC), ATP synthase, and NADP ⁺ reductase. Water molecules are shown entering at PSII and oxygen is shown leaving, but the	Distribute the diagram and reaction card set. Say: 'You are not receiving notes yet — you are going to build your own understanding by sequencing these events and placing them on the diagram. This is the same thing a biochemist does when they are working out a new pathway.' Allow 12 minutes of group work. Circulate actively — visit each	Sequencing and spatial mapping (card sort + diagram annotation): Physically manipulating sequence cards activates embodied cognition — learners must justify the ORDER of events, not just recall them. Matching cards to spatial locations on a diagram builds the mental model of the thylakoid as a molecular machine, which is the conceptual core of this	During circulation, check: (a) Do groups correctly place photolysis (water splitting) at PSII before the ETC? (b) Do groups distinguish between ATP (from ETC/chemiosmosis) and NADPH (from PSI)? (c) Can any learner explain WHY electrons must come from water? During gallery correction, count the number of groups with correct full sequences

for similar videos  EXERCISE: The light-dependent reactions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	intermediate steps are blank. Learners also receive a 'reaction card set' — 6 small cards each describing one event (e.g., 'Light strikes chlorophyll in PSII, exciting electrons to a higher energy level'; 'Water is split: $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$'; 'Electrons pass along the ETC, releasing energy used to pump H^+ ions'; 'H^+ ions flow back through ATP synthase, generating ATP'; 'Electrons reach PSI, are re-energised by light, and reduce NADP^+ to NADPH'; 'ATP and NADPH leave the thylakoid to power the next stage'). Working in groups of 3, learners sequence the cards, match each card to a location on the diagram, and annotate the diagram with the card text. Groups have 12 minutes for this activity.	group at least once. Use targeted questions rather than answers: 'Where do the electrons that travel along the ETC come from? Look at your card about water.' 'Your group has ATP synthase before the ETC — does that make sense if ATP synthase needs a H^+ gradient to work?' After 12 minutes, conduct a whole-class 'gallery correction': project the correct annotated diagram on the board step by step, one card at a time. For each step, cold-call a different group: 'Group 3 — Otieno, where did you place the photolysis card, and why?' Allow 10 seconds of WAIT TIME before accepting an answer. After each group response, ask the class: 'Does your group agree or disagree? Show thumbs.' Resolve disagreements by returning to evidence: 'What does the card say happens to the water molecules?'	lesson. The gallery correction provides immediate feedback without removing learner agency.	— target $\geq 75\%$ correct after group work. Record names of groups who misplace ATP synthase or confuse PSII/PSI for targeted follow-up in Phase 3.
Explain Phase  VIDEO: Conceptual overview of light dependent reactions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: The light-dependent reactions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners receive a graphic organiser: a horizontal cause-and-effect chain with 6 blank boxes and connecting arrows. The first box is pre-filled: 'LIGHT ENERGY strikes the leaf.' The last box is pre-filled: 'ATP and NADPH are available for the next stage of photosynthesis.' Learners individually complete the 4 intermediate boxes using their annotated diagrams. They then answer 3 written questions in their notebooks: (Q1) 'In the Elodea experiment, we measured oxygen bubble production as a proxy for photosynthesis rate. Using your diagram, explain exactly where that oxygen comes from and what would happen to bubble production if we added a chemical that blocks photolysis.' (Q2) 'The sukuma wiki under the mango	Say: 'You now have the annotated diagram in front of you. I want you to use it to build a cause-and-effect chain — this is a tool scientists use to show how one event triggers the next.' Distribute the graphic organiser. Give 6 minutes of SILENT individual work. After 6 minutes, say: 'Now look at the three questions on the board. These connect our molecular diagram back to the two phenomena we have been investigating all term. Write your answers in your notebook — full sentences, please.' Give 8 minutes for written responses. After 8 minutes, facilitate a class discussion for Q2 (the sukuma wiki shade question) — this is the most complex and directly ties to the driving question. Cold-call 3 learners (rotate to learners who	Cause-and-effect chaining + transfer questions: The graphic organiser forces learners to make the mechanistic logic explicit in their own words rather than copying notes. The three transfer questions escalate in cognitive demand — Q1 tests comprehension, Q2 requires application to the anchoring phenomenon (sukuma wiki), and Q3 requires evaluation/inference. This mirrors how scientists use mechanistic models to explain novel observations.	Collect notebooks at the end of Phase 3 OR use a rapid 'exit ticket' version: teacher reads Q2 aloud and asks learners to hold up 1 finger (can explain clearly), 2 fingers (partially understand), or 3 fingers (still confused). Target: $\geq 70\%$ show 1 finger. For learners showing 3 fingers, assign a peer-tutor from the 1-finger group for a 3-minute re-explanation before the DQB phase. In notebooks, look for: (a) correct identification of photolysis as the oxygen source; (b) explicit link between reduced light $\rightarrow$ less chlorophyll excitation $\rightarrow$ less ATP/NADPH $\rightarrow$ slower Calvin cycle $\rightarrow$ slower growth/pallor; (c) acknowledgement that Q3 requires distinguishing between light reactions and the use of their products.

	shade became pale and grew slowly over several weeks. Using what you now know about the light-dependent reactions, explain at the molecular level why shade would slow the plant's growth — even if water and CO₂ are available.' (Q3) 'The sukuma wiki recovers overnight even without sunlight. Does the light-dependent reaction run at night? What does this suggest about what the plant uses during the night to repair itself?'	have not yet spoken today): 'Zawadi, take us through your explanation for Q2.' After each response, ask: 'Can someone add a molecular detail Zawadi's explanation is missing?' For Q3, say: 'This is a tricky one. I am going to give you 15 seconds to think silently before anyone speaks.' [Allow 15 seconds of WAIT TIME.] Then cold-call: 'Kipchoge — what do you think?' Guide learners toward the idea that the plant uses pre-formed ATP and NADPH (or synthesised sugars from the Calvin cycle) stored during daylight for overnight repair, and that the light reactions themselves are DARK-independent only in the sense that no light reactions run — but the products are already stored.		
Driving Question Board (DQB) Creation  VIDEO: Photosynthesis: Overview of the light-dependent reactions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: The light-dependent reactions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The class DQB (established in Lesson 1) is displayed at the front of the room. It contains the anchoring driving question ('Why does our sukuma wiki wilt in the midday sun even when the soil is wet — and what does a plant actually need to stay green, grow, and make its own food?') and sticky-note question clusters from previous lessons. Learners individually write on a sticky note: (a) ONE thing they can now explain about the sukuma wiki phenomenon that they could NOT explain before today's lesson; (b) ONE new question that today's lesson has raised for them. They post their notes on the DQB in two columns: 'NOW EXPLAINED' and 'NEW QUESTIONS.' The teacher reads selected notes aloud and briefly connects them to the lesson sequence: which new questions will be addressed in Lessons 6–7	Point to the DQB and say: 'When we started this unit, we looked at the sukuma wiki in Amina's kitchen garden and it raised a lot of questions we could not answer. Let's see how far we have come.' Read the driving question aloud. Say: 'Think about what you knew on Day 1 versus what you know right now. On your sticky note, write one thing you can NOW explain — be specific, use the molecular language from today. And write one new question that today's lesson has opened up for you.' Give 3 minutes of individual silent writing. Invite learners to post notes on the DQB. Read 4–5 'NOW EXPLAINED' notes aloud, affirming precise language: 'Excellent — Wanjiku wrote: the oxygen in the bubbles comes from water molecules being split at Photosystem II, not from CO₂. That is a Nobel Prize-level insight — it	Public knowledge-building on the DQB: The DQB functions as a class-wide 'working model' of understanding — a living artefact that grows across the 10-lesson sequence. Making new understanding visible and public reinforces that science is cumulative and social. The two-column structure ('NOW EXPLAINED' vs. 'NEW QUESTIONS') models scientific practice: answering one question always generates new questions.	Teacher reads all sticky notes before the next lesson. Indicators of lesson success: 'NOW EXPLAINED' notes should include at least one of — photolysis as source of O₂, role of chlorophyll in absorbing light, ATP/NADPH as energy carriers, or connection to Elodea bubble data. 'NEW QUESTIONS' should include forward-pointing questions (e.g., 'What does ATP do next?' 'Where does the CO₂ go?') — these indicate learners understand that the light reactions are only PART of photosynthesis, which is the prerequisite understanding for Lesson 6.

	(Calvin cycle / light-independent reactions).	took scientists decades to prove that.' Read 2–3 'NEW QUESTIONS' notes and say: 'These questions about what happens to the ATP and NADPH next — those are exactly what we will investigate in Lesson 6, the Calvin cycle.' Do NOT resolve these questions — preserve productive curiosity.		
Model Building Phase  VIDEO: Photosynthesis: Overview of the light-dependent reactions - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: The light-dependent reactions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners retrieve their cumulative photosynthesis model begun in Lesson 3 (a hand-drawn diagram of the chloroplast that has been revised after each lesson). In Lesson 4, their models showed the chloroplast as a 'black box' with light and water entering and oxygen and 'energy' leaving. Today, learners open the black box by adding internal thylakoid structure to their model. Using their annotated diagram from Phase 2, they add: (a) a thylakoid stack (granum) with labelled PSII and PSI; (b) arrows showing the electron flow from water → PSII → ETC → PSI → NADP⁺/NADPH; (c) ATP synthase with a curved arrow showing H⁺ flow; (d) oxygen leaving the thylakoid; (e) ATP and NADPH arrows pointing toward the stroma (labelled 'to be used in the next stage'). Learners annotate their model with a key that includes: 'INPUTS to this stage: light energy, water, ADP+Pi, NADP⁺' and 'OUTPUTS of this stage: ATP, NADPH, O₂.' They write a 2-sentence 'model update note' at the bottom: 'I added ____ to my model today because ____.' This helps explain the sukuma wiki phenomenon because ____.'	Say: 'Get out your photosynthesis model from Lesson 3. Look at it — you drew the chloroplast as a box. You wrote light and water going in, and oxygen and energy coming out. Today you are going to show WHAT IS HAPPENING INSIDE that box.' Display your own model (teacher's reference copy, projected) showing the Lesson 3–4 version with the thylakoid still unlabelled. Say: 'I want you to add the internal machinery we traced today. Use your annotated diagram as a reference — do not copy from a textbook, use YOUR diagram.' Give 8 minutes for model revision. Circulate and ask 3 targeted questions to individual learners: 'Show me on your model where the oxygen is produced — can you point to the exact step?' 'Your model shows ATP being made — can you write next to it why ATP is needed? What will the plant use it for?' 'Looking at your model, if chlorophyll is destroyed by a disease that bleaches leaves — like what happened to the shaded sukuma wiki — which step would stop first?' After 8 minutes, pairs do a 'model swap': each learner reviews their partner's model and writes one specific 'star' (something correct and precise) and one 'question' (something unclear or missing). Partners	Iterative model revision with peer critique: The cumulative model is the learner's personal scientific artefact — it grows in explanatory power across the unit, mirroring how scientific models develop through cycles of evidence and revision. The 'star and question' peer review protocol develops scientific communication skills and ensures learners evaluate evidence claims, not just copy information. The 'model update note' requires learners to explicitly connect the model revision to the phenomenon, preventing the model from becoming an abstract exercise.	Teacher collects models (or photographs them) at the end of the lesson. Evaluate against a 4-point rubric: (1) Thylakoid structure present with PSII and PSI labelled; (2) Electron flow correctly sequenced from water to NADPH; (3) ATP synthase correctly placed with H⁺ gradient indicated; (4) Model update note explicitly connects O₂ production or ATP/NADPH to the sukuma wiki phenomenon. Target: ≥ 80% of learners achieve 3/4 or higher. Learners scoring 1/4 are flagged for a 5-minute re-teach at the start of Lesson 6 using a simplified two-box model (PSII box and PSI box with arrows).

		discuss for 2 minutes. Close by saying: 'Your model is not finished — it will never be finished, because science is never finished. In Lesson 6 we will add the Calvin cycle to the stroma section. By the end of this unit your model will explain everything we see in Amina's kitchen garden.'	
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your notes before Lesson 6:

1. PHENOMENON CONNECTION: During Phase 3, when learners answered Q2 (shade → pale sukuma wiki), what percentage could articulate the molecular chain — reduced light → less chlorophyll excitation → less ATP/NADPH → slower growth? If fewer than 70% could do this, plan a 5-minute phenomenon re-anchor at the start of Lesson 6.
2. MISCONCEPTION WATCH: Did any learners claim that oxygen comes from CO₂ (a very common misconception)? If yes, plan a brief demonstration in Lesson 6 using the historical context of van Niel's experiment with purple sulfur bacteria — a powerful misconception-buster.
3. CARD SORT ENGAGEMENT: Were all group members contributing to the card sort, or were dominant learners taking over? Consider assigning roles (Reader, Placer, Challenger, Reporter) in Lesson 6 group activities.
4. DQB QUALITY: Were the 'NEW QUESTIONS' on the DQB genuinely forward-pointing toward the Calvin cycle (e.g., 'What does ATP do next?', 'Where does CO₂ enter?')? If learners are not yet asking about the next stage, the bridge to Lesson 6 (light-independent reactions) may need more scaffolding.
5. MODEL DEPTH: Did learners' model update notes connect to the sukuma wiki phenomenon with molecular specificity, or were the connections vague (e.g., 'this helps the plant grow')? Use the strongest model update notes as exemplars at the start of Lesson 6.
6. PACING: The Phase 2 card sort (12 minutes) and Phase 3 transfer questions (14 minutes) are the most time-intensive segments. If either ran over time, consider whether the gallery correction in Phase 2 can be shortened by having groups photograph their card arrangements and share one image per group rather than presenting verbally.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners reviewed Elodea oxygen-bubble count data from Lesson 4 and examined a half-labelled thylakoid membrane diagram with a set of 6 reaction-event cards describing the steps of the light-dependent reactions.
What did I learn?	Learners traced the sequence of light-dependent reactions: light absorption by chlorophyll in PSII and PSI → excitation and transfer of electrons → photolysis of water ($2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$) as the electron source → chemiosmotic synthesis of ATP (photophosphorylation) via ATP synthase → reduction of NADP ⁺ to NADPH at PSI → release of ATP and NADPH to the stroma for use in the Calvin cycle. They connected O ₂ release to the Elodea bubble observations and connected reduced ATP/NADPH production under shade to the pale, slow-growing sukuma wiki.
How does this explain the phenomenon?	More light hitting the sukuma wiki leaf means more chlorophyll molecules are excited in Photosystem II, driving faster photolysis of water, faster electron flow along the ETC, and greater production of ATP and NADPH — which is why the Elodea produces more bubbles at higher light intensities. When the sukuma wiki is moved into the shade of the mango tree, the reduced light energy means less chlorophyll excitation, slower photolysis, and therefore less ATP and NADPH produced — starving the Calvin cycle of the energy and reducing power it needs to fix CO ₂ into sugars, which explains the slow growth and yellowing (pallor) of the shaded

	plant.
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LESSON 6 (80 minutes): Light-Independent Reactions: How the Calvin Cycle Builds Glucose from CO₂

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to explain the light-independent reactions (Calvin cycle) of photosynthesis, including the fixation of CO ₂ and the role of ATP and NADPH in producing glucose (G3P), and to connect both stages of photosynthesis in a unified flow diagram.
Knowledge	Learners will know that: (1) the light-independent reactions (Calvin cycle) occur in the stroma of the chloroplast; (2) CO ₂ from the air is fixed by the enzyme RuBisCO onto a 5-carbon acceptor molecule (RuBP); (3) ATP and NADPH produced in the light-dependent stage are used to reduce the 3-carbon intermediate (3-PGA) to glyceraldehyde-3-phosphate (G3P); (4) G3P is used to synthesise glucose and to regenerate RuBP; (5) without sufficient light, less ATP and NADPH are available, slowing G3P and glucose production.
Skills	Learners will be able to: (1) construct a simplified flow diagram linking the light-dependent and light-independent reactions; (2) use evidence from the diagram to explain why shaded sukuma wiki grows more slowly; (3) interpret and annotate diagrams showing inputs and outputs of the Calvin cycle; (4) apply the concept of limiting factors to predict outcomes when ATP/NADPH supply is reduced.
Attitudes	Learners will appreciate the elegance of how a plant converts simple inorganic molecules — CO ₂ , water, and light energy — into the organic matter that feeds nearly all life, including the sukuma wiki on Kenyan dinner tables. Learners will value evidence-based reasoning when revising their earlier predictions about shade and plant growth.
Key Inquiry Question	How do plants make their own food? What factors affect the rate at which plants make food?
Purpose in Storyline	Lesson 6 is the EXPLAIN phase for the light-independent reactions. Having established in Lessons 4–5 that light drives the splitting of water and the production of ATP and NADPH, learners now discover what the plant does with those energy carriers. They trace the journey of a CO ₂ molecule from the air through RuBP fixation to G3P and finally to glucose. By constructing a two-stage flow diagram they can, for the first time, give a complete biochemical explanation for why shaded sukuma wiki in the family's kitchen garden — though protected from wilting — gradually turns pale and grows slowly: less light → less ATP and NADPH → Calvin cycle slows → less glucose → slower growth and eventual chlorophyll breakdown.
Safety Notes	No hazardous chemicals or equipment are used in this lesson. If printed card-sort materials are laminated, ensure scissors (if used to cut un-laminated sets) are handled safely. Learners should remain seated during model-building to avoid crowding. If the lesson is conducted in a laboratory, standard lab rules apply: no eating or drinking, bags stored under benches.

B. LESSON OVERVIEW

In Lesson 6, learners shift from the light-dependent reactions (covered in Lesson 5) to the light-independent reactions — the Calvin cycle. The lesson opens with a Predict phase in which learners hypothesise what happens to the ATP and NADPH produced in the light stage, grounding their thinking in the sukuma wiki phenomenon. The Observe phase uses an annotated diagram walkthrough combined with a structured card-sort activity in which learners physically sequence the steps of the Calvin cycle using locally labelled cards (CO₂ from Nairobi air, RuBP, 3-PGA, G3P, glucose → ugali/sukuma wiki biomass). The Explain phase uses a Think-Pair-Share and guided Socratic questioning to build a two-stage flow diagram on the board collaboratively. In the DQB phase, learners revisit their posted questions and answer at least two with new evidence. The Model-Building phase asks learners to revise their personal photosynthesis models to include both stages, with arrows showing how reduced light supply in shade throttles the whole system. Formative assessment is embedded throughout via cold-calling, whiteboards, and an exit card.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Photosynthesis: Calvin cycle Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Calvin Cycle (Flexbook) Source: CK-12  Search ARES for similar readings	Learners silently read a short scenario card (printed or displayed): 'Amina's sukuma wiki under the mango tree receives only 20% of the sunlight it would get in the open garden. Her friend tells her the plant cannot make enough food. Amina asks: what exactly runs out first — and why does the plant grow slowly rather than just wilting?' Learners individually write a 3-sentence prediction in their exercise books answering: (a) What do you think happens to the ATP and NADPH made in the light reactions when the plant is in shade? (b) Where in the chloroplast do you think the next stage of photosynthesis happens? (c) What molecule do you predict is the first product the plant builds toward making glucose? Learners then share predictions with a seat partner before the teacher cold-calls.	Display the scenario card and read it aloud expressively. Say: 'We have spent two lessons learning how light energy is captured and converted into ATP and NADPH. Today we follow that energy further into the chloroplast. Before I show you anything new, I want to know what YOU already think.' Circulate for 3 minutes while learners write — resist answering questions; respond to uncertainty with 'Write down your best guess; that is the point.' After 2 minutes of pair discussion, cold-call 3 learners (choose one high-confidence, one mid, one low based on prior lesson observations). For each response say: 'Thank you — keep that prediction. We will come back to it.' Write the three predictions verbatim on a corner of the board under the heading 'Our Predictions — Lesson 6'. Do NOT confirm or correct yet. WAIT TIME: allow 10 seconds of silence after each cold-call prompt before redirecting.	Predict-Before-Instruction (anticipatory activation): By committing a prediction to paper before instruction, learners surface and make explicit their current mental models of what ATP and NADPH are 'for'. This creates cognitive dissonance when the actual Calvin cycle mechanism is revealed, deepening encoding. Connecting predictions to the named phenomenon (Amina's shaded sukuma wiki) ensures the question feels authentic rather than abstract.	Teacher circulates and scans written predictions. Observable indicators of prior understanding: (1) Learner correctly names ATP/NADPH as energy/hydrogen carriers for the next stage — indicates strong carry-over from Lesson 5; (2) Learner mentions 'stroma' — indicates recall of chloroplast structure from Lesson 4; (3) Learner cannot move beyond 'the plant uses energy to grow' — flags need for scaffolding during Observe phase. Teacher mentally notes 2–3 learners who need targeted questioning during the card-sort.
Observe Phase  VIDEO: Photosynthesis: Calvin cycle Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Calvin Cycle Practice Source: CK-12  Search ARES	Part A — Annotated Diagram Walk (10 min): Each learner receives (or views on screen) a simplified diagram of the Calvin cycle showing: $\text{RuBP (5C)} + \text{CO}_2 \rightarrow 6 \times 3\text{-PGA (3C)}$ $[\text{RuBisCO}] \rightarrow 6 \times \text{G3P (3C)}$ $[\text{uses ATP} + \text{NADPH}] \rightarrow 1 \text{ G3P exits to make glucose} / 5 \text{ G3P regenerate RuBP} [\text{uses ATP}]$. Key labels are blanked out. Learners use a coloured pencil to trace the path of (i) carbon atoms — in red; (ii) ATP use — in blue; (iii) NADPH use — in green. They annotate where in the chloroplast	Distribute diagram sheets and say: 'You have a blank Calvin cycle diagram. Using THREE colours, I want you to trace three different stories happening at the same time: the carbon story in RED, the ATP story in BLUE, the NADPH story in GREEN. You have 8 minutes. Go.' Set a visible timer. After 5 minutes, pause the class: 'Raise your hand if you have traced where ATP is used — look at your neighbour's diagram and see if you agree on the same step.' Allow 90 seconds of paired	Colour-Coded Tracing + Physical Card-Sort (Embodied Cognition + Sequential Reasoning): The colour-tracing makes abstract molecular flows visible and distinguishable, reducing cognitive overload. The physical manipulation of cards (embodied cognition) helps learners internalise the cyclic, regenerative nature of the Calvin cycle more durably than reading alone. Embedding a Kenya-specific card ('CO ₂ enters leaf stomata above Lake Naivasha farmland') and the	Teacher checks each group's card sequence for two critical placements: (1) 'ATP and NADPH from light reactions arrive in stroma' must precede 'G3P produced' — if groups place it after, this reveals a misconception that the Calvin cycle generates its own energy; (2) The last phenomenon-link card must be placed as a consequence of reduced ATP/NADPH, not as a separate standalone fact. Teacher notes groups with incorrect sequencing for targeted support in

for similar readings	(stroma) this occurs. Part B — Card-Sort Sequencing (12 min): In groups of 3–4, learners receive a set of 10 laminated cards. Cards include: 'CO₂ enters leaf stomata above Lake Naivasha farmland', 'CO₂ fixed onto RuBP by RuBisCO enzyme', '6 molecules of 3-PGA formed', 'ATP and NADPH from light reactions arrive in stroma', '6 molecules of G3P produced', '1 G3P used to build glucose → sukuma wiki leaf cell wall and sugars', '5 G3P used to regenerate RuBP', 'RuBP ready to fix another CO₂', 'Cycle repeats 6 times to net one glucose molecule', 'Less light → less ATP/NADPH → cycle slows → less glucose → slower growth'. Groups sequence the cards in a cycle layout on their desk and sketch the arrangement in their notebooks.	checking. Then introduce the card-sort: 'Now your group will build the Calvin cycle as a physical sequence. Every card must be placed — there are no trick cards. The last card is your most important one: connect it back to Amina's shaded sukuma wiki.' Circulate to each group — do NOT give answers; instead prompt with: 'Where does the CO₂ come from before it enters the leaf?' and 'If this step uses ATP, what lesson told us where that ATP was made?' Cold-call 2 groups to justify placement of the card 'Less light → less ATP/NADPH → cycle slows'. WAIT TIME: 12 seconds after each group prompt.	phenomenon-linked card ('less light → slower growth') anchors biochemistry in the lived context of the driving question.	Explain phase. If >50% of groups misplace the ATP/NADPH card, teacher pauses the class for a brief whole-group clarification before proceeding.
Explain Phase  VIDEO: Photosynthesis: Calvin cycle - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: The Calvin cycle Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Part A — Collaborative Diagram Construction (14 min): Whole class co-constructs a two-stage photosynthesis flow diagram on the board. Starting with an empty template (two large boxes labelled 'LIGHT-DEPENDENT REACTIONS — Thylakoid Membrane' and 'LIGHT-INDEPENDENT REACTIONS — Stroma'), learners are cold-called to come forward and add: inputs (light, water, CO₂, ADP+Pi, NADP⁺), outputs (O₂, ATP, NADPH, G3P, glucose), and connecting arrows labelled with molecule names. Each contributor must say one sentence explaining their addition. Part B — Think-Pair-Share Explanation (8 min): Learners answer in writing: 'Amina's sukuma wiki under the mango tree receives less light. Use the two-stage flow diagram to	Begin by saying: 'We are now going to build the complete story of photosynthesis together — both stages — on the board. This diagram should explain everything we have observed about Amina's sukuma wiki.' Point to the empty two-box template and say: 'Who can tell me the first input into the light-dependent stage?' Cold-call (do not take volunteers first) — 10 seconds WAIT TIME. As learners add components, narrate connections: 'Notice — the ATP made HERE in the thylakoid travels THERE into the stroma. The Calvin cycle does not make its own energy; it borrows energy produced by light.' Use a red marker to draw a bold arrow from Box 1 to Box 2 labelled 'ATP + NADPH' and say: 'This arrow is the bridge. If light weakens, this arrow gets thinner. What happens	Collaborative Board Construction + Causal Chain Narration (Visible Thinking + Mechanistic Reasoning): Co-constructing the diagram makes the class's collective understanding visible and correctable in real time — errors are caught publicly and revised, modelling scientific practice. The explicit causal chain narration (using the connector 'therefore' between each step) trains learners in mechanistic explanation, a core NGSS Science and Engineering Practice (Constructing Explanations). Grounding every step in the sukuma wiki phenomenon ensures explanation remains purposeful rather than rote.	Observable indicators: (1) During board construction — learner correctly places 'ATP + NADPH' as outputs of Box 1 AND inputs of Box 2 without prompting: strong understanding of inter-stage dependency; (2) Written Think-Pair-Share — learner's causal chain includes all six required terms in a logical sequence: meets learning outcome; (3) Learner conflates light reactions and Calvin cycle as happening in the same location (thylakoid): misconception flagged — teacher immediately clarifies using the diagram. Teacher collects 5 randomly selected written responses to review after class.

	explain, step by step, why the plant grows more slowly. Your answer must mention: ATP, NADPH, Calvin cycle, G3P, glucose, and growth.' Pairs compare answers and refine. One pair is cold-called to read their explanation aloud.	downstream?' Cold-call 2 learners. For the Think-Pair-Share, circulate and read written responses — prompt weaker responses with: 'You said the plant grows slowly — but WHY? What molecule is missing that the plant needs to build new cells?' After pairs share, narrate the complete causal chain on the board: Less light → fewer electrons excited → less ATP + NADPH produced → Calvin cycle cannot reduce 3-PGA to G3P efficiently → less G3P → less glucose synthesised → less energy and fewer carbon skeletons for cell growth → sukuma wiki grows slowly and eventually loses chlorophyll. WAIT TIME: 12 seconds after every cold-call prompt.		
Driving Question Board (DQB) Creation  VIDEO: Photosynthesis <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Calvin Cycle (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board. Each learner reviews the sticky notes posted in Lessons 1–5. They identify at least TWO questions on the board that they can now answer using knowledge from Lessons 5–6 (light-dependent and light-independent reactions). They write a 2–3 sentence answer on a differently coloured sticky note and attach it below the original question. They also add ONE new question prompted by today's lesson (e.g., 'If the plant makes G3P, how exactly is G3P turned into the cellulose of the sukuma wiki leaf?' or 'Does the Calvin cycle stop completely at night, or just slow down?'). Whole class briefly shares two answered questions and one new question.	Direct learners to the DQB: 'Look at the questions we have built up over six lessons. You now know about both stages of photosynthesis. Pick any two questions you can answer today — not questions about transport or wilting yet, those are coming. Write your answer on a GREEN sticky note and pin it below the question.' Allow 5 minutes. Then say: 'Before we leave the DQB, add one new question the Calvin cycle has raised for you — something you are still curious about. Pink sticky note, top right cluster.' After posting, cold-call 2 learners to read an answered question and their evidence. Say: 'Notice how our board is filling up with answers — we are building a scientific explanation together, piece by piece, just like researchers at KEMRI or the University of Nairobi would do.' WAIT TIME: 10 seconds when asking for new	Driving Question Board Revisit (Metacognitive Monitoring + Collaborative Knowledge Building): Returning to the DQB makes learning progress visible — learners literally see their questions being answered, which builds epistemic confidence. Adding new questions models the scientific disposition that answering one question generates further inquiry. Referencing Kenyan research institutions (KEMRI, University of Nairobi) contextualises the practice as authentically scientific and locally relevant.	Teacher scans green sticky-note answers for accuracy and causal depth. Observable indicators: (1) Learner answers 'Why does shade cause slow growth?' with a complete causal chain mentioning ATP/NADPH and glucose: demonstrates integration of Lessons 5–6; (2) Learner's answer is limited to 'less light = less food': indicates surface-level understanding — teacher makes a note to probe this learner during model-building; (3) Quality of new questions: questions about the biochemical fate of G3P or the role of enzymes indicate higher-order curiosity beyond curriculum floor; questions about unrelated topics indicate the learner may be confused about lesson boundaries.

		questions.		
Model Building Phase  VIDEO: Breaking down photosynthesis stages <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Calvin Cycle Practice Source: CK-12  Search ARES for similar readings	Learners open their personal photosynthesis model (started in Lesson 3 and revised in Lessons 4–5). They now add a second chamber to their chloroplast diagram labelled 'Stroma — Calvin Cycle'. Inside this chamber they draw and label: CO₂ entering, RuBP, 3-PGA, G3P, glucose arrow exiting, and the ATP + NADPH arrows arriving from the thylakoid (their previous model). They add a 'dimmer switch' arrow from the sun to the light reactions, with a note: 'Less sunlight under mango tree → dimmer switch reduces ATP/NADPH flow → Calvin cycle produces less G3P → less glucose → slow growth.' They annotate their model with one sentence explaining the sukuma wiki phenomenon at the biochemical level. A volunteer shares their revised model under the document camera (or holds it up) for class feedback.	Say: 'Open your photosynthesis model. It currently shows the light reactions. Today you are adding the second engine — the Calvin cycle — and connecting the two. Draw the stroma chamber next to your thylakoid. Use your card-sort sequence and the board diagram as references.' Allow 8 minutes of individual drawing. Circulate and check that: (1) ATP/NADPH arrows connect thylakoid to stroma correctly; (2) CO₂ is shown entering the stroma (not the thylakoid); (3) the 'dimmer switch' metaphor is present. Prompt learners who are stuck: 'What comes OUT of the light reactions that the Calvin cycle needs? Draw that first.' After 8 minutes, invite one volunteer to share under the document camera. Guide class feedback with: 'What did [name] include that strengthens the model? What one thing would make it even more accurate?' WAIT TIME: 10 seconds after the feedback prompt. Close by saying: 'Your model can now explain why sukuma wiki under a mango tree grows slowly at the molecular level. Next lesson we will use this model to explore the rate of photosynthesis and what happens when CO₂ or temperature changes.'	Cumulative Model Revision (Progressive Model-Based Reasoning): Requiring learners to physically add to and revise an existing model (rather than draw a new one each lesson) reinforces that scientific models are living, improving representations — a core NGSS practice (Developing and Using Models). The 'dimmer switch' metaphor scaffolds systems thinking by making the regulatory relationship between light intensity and downstream chemistry concrete and memorable. Peer feedback during the share cultivates the scientific practice of critique and revision.	Exit Card (last 3 minutes): Teacher distributes a half-page card with two prompts: (1) 'Draw a simple arrow diagram showing how ATP made in the thylakoid is used in the stroma. Label the inputs and outputs of each stage.' (2) 'In one sentence, explain why Amina's sukuma wiki under the mango tree grows slowly — use the words ATP, Calvin cycle, and glucose.' Teacher collects all exit cards. Observable indicators of mastery: Prompt 1 — correct bidirectional connection between stages with ATP/NADPH labelled on the connecting arrow; Prompt 2 — causal sentence includes all three required terms in a logical sequence. Cards with missing connections or absent terms are flagged for reteaching at the start of Lesson 7.

D. TEACHER REFLECTION

After the lesson, consider: (1) Were learners able to articulate the causal chain — less light → less ATP/NADPH → slower Calvin cycle → less glucose → slower growth — without prompting? If not, the 'bridge arrow' metaphor may need more emphasis in Lesson 7's recap. (2) Did the card-sort reveal any persistent misconception that the Calvin cycle produces its own ATP independently of the light reactions? If so, plan a 3-minute 'energy budget' mini-activity at the start of Lesson 7. (3) How many exit cards showed complete two-stage diagrams with correctly labelled connecting arrows? Set a target of ≥70% of learners achieving both exit card prompts correctly. (4) Were the DQB new questions sophisticated enough to suggest learners are thinking beyond the curriculum minimum — e.g., asking about enzyme kinetics, the fate of

G3P, or night-time Calvin cycle activity? Use the highest-quality new DQB questions to motivate the limiting-factors discussion in Lesson 7. (5) Did references to Kenyan contexts (Naivasha farmland, sukuma wiki biomass, KEMRI) land as meaningful or superficial? Adjust specificity of local references for Lesson 7 if learners seemed disengaged by them.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners traced the colour-coded Calvin cycle diagram (carbon in red, ATP use in blue, NADPH use in green) and physically sequenced 10 laminated cards — including 'CO ₂ fixed onto RuBP by RuBisCO enzyme', 'ATP and NADPH from light reactions arrive in stroma', and 'Less light → less ATP/NADPH → cycle slows → less glucose → slower growth' — to reconstruct the full light-independent reaction pathway in the context of Amina's shaded sukuma wiki.
What did I learn?	Learners learned that the light-independent reactions (Calvin cycle) occur in the stroma of the chloroplast, where CO ₂ is fixed onto RuBP by RuBisCO to form 3-PGA, which is then reduced to G3P using ATP and NADPH imported from the light-dependent reactions; G3P is used to synthesise glucose and to regenerate RuBP. Less light reduces ATP and NADPH production, slowing the Calvin cycle and producing less glucose, which explains why shaded sukuma wiki grows more slowly despite not wilting.
How does this explain the phenomenon?	Using a collaboratively constructed two-stage flow diagram, learners explained the complete causal chain linking reduced sunlight under the mango tree to slow growth in sukuma wiki: less light → fewer electrons excited in thylakoid → less ATP and NADPH produced → Calvin cycle cannot efficiently reduce 3-PGA to G3P → less glucose synthesised → fewer carbon skeletons and less energy for cell division and growth → sukuma wiki grows slowly and eventually loses chlorophyll pigmentation.

LESSON 7 (80 minutes (double lesson)): Factors Affecting the Rate of Photosynthesis — Planning & Carrying Out an Investigation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To plan and carry out a quantitative investigation into how light intensity affects the rate of photosynthesis, using oxygen bubble production as a measurable indicator.
Knowledge	Learners will know that: (1) the rate of photosynthesis can be measured indirectly by counting oxygen bubbles produced per unit time; (2) light intensity is a limiting factor — increasing it increases the rate up to a maximum; (3) other factors (CO ₂ concentration, temperature) can also limit the rate; (4) a limiting factor is the factor present at the lowest/most restrictive level that controls the overall rate of a process.
Skills	Learners will be able to: (1) formulate a testable hypothesis about light intensity and photosynthesis rate; (2) identify and control variables (independent, dependent, controlled); (3) collect quantitative data by counting oxygen bubbles per minute at varying light intensities; (4) record data in a correctly labelled table; (5) plot a line graph of rate of photosynthesis against light intensity; (6) identify the limiting factor from graph shape and data trends.
Attitudes	Learners will: (1) appreciate that careful, systematic data collection leads to reliable scientific conclusions; (2) show curiosity and persistence when results do not immediately match predictions; (3) value collaborative work and fair division of roles during an investigation.
Key Inquiry Question	What factors affect the rate at which plants make food?
Purpose in Storyline	In Lesson 6, learners established that light is a raw material/energy source for photosynthesis and began to explain why the shaded sukuma wiki turned pale. Lesson 7 now moves from qualitative explanation to quantitative evidence: learners directly measure HOW MUCH light intensity matters by generating and analysing their own data. This advances the driving question — the wilting sukuma wiki recovers at night partly because metabolic demand drops, but the shade problem is fundamentally about insufficient light intensity for adequate photosynthesis. By the end of this lesson, learners will have experimental data showing a rate–intensity relationship, bringing them closer to explaining both the pallor (low light → low photosynthesis rate → less glucose, less chlorophyll synthesis) and the wilting (high light → high transpiration rate — a connection to flag for Lesson 10).
Safety Notes	– If using a lamp, warn learners not to touch the bulb — it becomes very hot. Keep water containers away from electrical sockets. • If using a live aquatic plant (e.g., Elodea or a locally available pond weed from Lake Victoria or a Nairobi park pond), handle specimens with care and wash hands after. • If sodium bicarbonate solution is used, avoid contact with eyes; learners should wear safety goggles. • Remind learners to report any broken glassware immediately. • If the lesson uses a simulated/printed dataset instead of live plants, no additional safety precautions are needed.

B. LESSON OVERVIEW

Learners begin by revisiting the phenomenon — the sukuma wiki turning pale under shade — and predict what will happen to photosynthesis rate as light intensity changes. They then plan a fair-test investigation (identifying variables, writing a hypothesis, designing a results table) before observing: either counting oxygen bubbles from an aquatic plant at different lamp distances, or analysing a teacher-provided authentic dataset from a similar experiment. They record data, plot line graphs, and identify the concept of a limiting factor. The lesson closes with learners updating the Driving Question Board and revising their class photosynthesis model to include the rate–intensity relationship.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Dependent & independent variables Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Independent versus dependent variables Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners are shown two photographs side by side on the board (or described verbally): (A) a dense sukuma wiki plot in full Nairobi midday sun — leaves bright green and upright in the morning; (B) the same plot after two weeks under a mango tree — leaves pale yellowish-green. The teacher poses the question: 'We already know light is needed for photosynthesis. Today I want you to think more precisely — if we give the plant MORE light, does photosynthesis speed up forever, or does it hit a ceiling? Draw a quick sketch-graph: put light intensity on the x-axis and rate of photosynthesis on the y-axis. Draw the curve you predict.' Learners work individually for 3 minutes, sketch their prediction graph in their exercise books, and write a one-sentence justification below it. Pairs then compare graphs for 2 minutes and note any differences.	1. Display or describe the two sukuma wiki images. Say: 'Cast your mind back to our phenomenon. The student's plants under the mango tree are pale and growing slowly. Last lesson we said light intensity is the key difference. Today we are going to be scientists — we will measure exactly how light intensity changes the rate of photosynthesis.' 2. Say: 'Before we collect ANY data, I want your prediction. Open your books. Draw x-axis: Light Intensity. Y-axis: Rate of Photosynthesis. Sketch the shape of the curve you expect. You have 3 minutes — go.' 3. WAIT — circulate silently for 3 minutes, noting 3–4 different graph shapes drawn by learners (flat line, straight diagonal, curve that levels off, curve that drops at high intensity). 4. After 3 minutes, cold-call 4 learners to hold up or describe their graphs. Ask each: 'Why did you draw it that shape?' Allow 10–15 seconds of WAIT TIME after each question before accepting a response. 5. Do NOT correct predictions yet. Say: 'Excellent — we now have four different hypotheses on the board. By the end of this lesson, our data will tell us which of these is closest to reality.'	Individual Prediction Sketch-Graphs — each learner externalises their mental model before instruction. Comparing different prediction shapes creates productive cognitive dissonance and motivates the investigation. The teacher logging 3–4 different shapes on the board transforms private predictions into a public 'class hypothesis set' that the data will adjudicate.	Observable indicator: Every learner has a sketch graph in their book with labelled axes and a written justification before the class moves on. Teacher scans books during circulation. Red flag: learners who draw no curve or cannot label axes — teacher provides a quick scaffold ('What are we changing? That goes on the x-axis. What are we measuring? That goes on the y-axis:').
Observe Phase  VIDEO: Dependent & independent	PART A — PLANNING (20 min): In groups of 4, learners receive an Investigation Planning Sheet. They must agree on and write: (1) Hypothesis (If light intensity	PART A: 1. Distribute Investigation Planning Sheets. Say: 'As scientists, we never just start collecting data. We plan. Your group has 8 minutes to	Plan-Before-You-Do Investigation Protocol — requiring learners to articulate variables and design a results table before touching equipment forces them to engage	Observable indicators: (1) Every group's planning sheet has all five components completed before data collection begins — teacher checks this as a gate. (2) Results

variables: graphing Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Independent versus dependent variables Source: Khan Academy (English - US curriculum) Search ARES for similar readings	increases, then the rate of photosynthesis will... because...); (2) Independent variable (light intensity — changed by moving the lamp to distances of 5 cm, 10 cm, 20 cm, 30 cm, 40 cm from the plant, OR by layering 1–4 sheets of muslin cloth over the lamp); (3) Dependent variable (number of oxygen bubbles produced per minute); (4) Controlled variables (at least 3: same plant shoot, same volume and temperature of water, same sodium bicarbonate concentration, same time interval counted); (5) Results table with correct headings and units. Groups share their draft tables with the adjacent group for peer review before data collection begins. PART B — DATA COLLECTION (25 min): OPTION 1 (Live plant): Each group sets up a beaker of sodium bicarbonate solution, places an aquatic plant shoot (Elodea or local pond weed) in a test tube submerged in the beaker, positions a lamp at the first distance, waits 2 minutes for the plant to equilibrate, then counts bubbles for 1 minute. Repeat for each distance (three trials per distance if time allows, otherwise one trial with class-pooled data). Record in the results table. OPTION 2 (Simulated dataset — printed card): Teacher distributes a printed authentic dataset card showing bubble counts at five light intensities from a Kenyan secondary school experiment. Learners read and record the data into their own results tables, then proceed identically to Option 1 groups.	complete the five planning steps on this sheet. I will be checking that every group can tell me: what are you changing, what are you measuring, and what are you keeping the same.' 2. After 5 minutes, stop the class. Cold-call one group: 'Group 3 — tell me your dependent variable and how you will measure it.' WAIT 10 seconds. If incomplete, prompt: 'How will you actually count photosynthesis happening? What can you see coming out of the plant?' 3. Before groups begin data collection, do a 3-minute peer-review swap of results tables. Say: 'Pass your planning sheet to the group behind you. Check: does the table have headings? Units? At least five different light intensities listed? Give it back with one written comment.' PART B: 4. Demonstrate the setup (or walk through the dataset card) for 3 minutes. Say: 'The bubbles you see — or the numbers on this card — represent oxygen being released. Every bubble is evidence of photosynthesis happening. More bubbles per minute = faster rate.' 5. Circulate continuously. At each group, ask one probing question: 'Why are we using sodium bicarbonate in the water?' (Answer: to ensure CO₂ is not the limiting factor — we want to isolate light.) WAIT 10 seconds for a response before intervening. 6. If a group finishes early: 'Calculate the average bubbles per minute across your three trials for	cognitively with experimental design (NGSS SEP 3: Planning and Carrying Out Investigations). The peer-review of tables builds metacognitive awareness of data quality. Connecting each bubble to the chemical equation of photosynthesis (O₂ as a product) anchors the measurement in the mechanistic understanding built in previous lessons.	tables have correct headings and units (Light Intensity / cm or arbitrary units; Rate of Photosynthesis / bubbles min⁻¹). (3) Graphs have labelled axes with units, a title, and a smooth curve. Red flags: groups using ruler-and-dot graphs instead of a best-fit curve (teacher intervenes: 'In biology, we draw the trend, not every dot — draw through the middle of the points'); learners with no units on axes.
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	PART C — GRAPH PLOTTING (10 min): Learners plot a line graph on grid paper: x-axis = Light Intensity (represented as $1/\text{distance}^2$ or simply as 'lamp distance in cm, with a note that closer = higher intensity'), y-axis = Rate of Photosynthesis (bubbles per minute). Learners draw a smooth best-fit curve (not dot-to-dot). They label the axes with units and give the graph a title.	each distance and add an Average column to your table.' PART C: 7. Say: 'Now plot your graph. Remember — smooth best-fit curve, not a zigzag. Title: Rate of Photosynthesis vs Light Intensity in [aquatic plant name].' 8. After 8 minutes, display a correctly drawn model graph on the board (without values). Ask: 'What shape is this curve? What does the flattening tell us?'		
Explain Phase  VIDEO: Dependent & independent variables Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Making appropriate scatter plots <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners analyse their graphs using three guiding questions printed on the board: (Q1) 'Describe the shape of your graph in one sentence.' (Q2) 'At low light intensities, what is happening to the rate as intensity increases? What does this tell you about light as a factor?' (Q3) 'At high light intensities, the curve flattens — the rate stops increasing even if we add more light. What might be limiting the rate now?' Groups discuss for 5 minutes, then the teacher introduces the formal concept of a LIMITING FACTOR: 'A limiting factor is the factor present at the lowest level that controls the rate of a process. When light is scarce, light is the limiting factor. When light is abundant, something else — CO₂ concentration or temperature — becomes the limiting factor.' Learners then complete a 'Claim–Evidence–Reasoning' (CER) frame individually: Claim — state whether light intensity affects rate of photosynthesis; Evidence — cite two specific data points from their table; Reasoning — explain using the limiting factor concept.	1. After graphs are plotted, say: 'Do not close your books. I want you to be a graph detective. Look at your curve and answer the three questions on the board in your group. You have 5 minutes.' WAIT — circulate, listening for use of the word 'increase', 'level off', 'maximum'. 2. After 5 minutes, cold-call three different groups for Q1, Q2, Q3 respectively. For Q3, if no group mentions CO₂ or temperature, ask: 'If adding more light doesn't help after a certain point, what else might the plant need that it's running low on?' WAIT 12 seconds. 3. Introduce the term LIMITING FACTOR using the board. Write: 'The factor present at the lowest/most restrictive level that controls the overall rate.' Say: 'Scientists in Kenya studying tea yield in Kericho use this concept — if there is enough rain but no fertiliser, fertiliser is the limiting factor for tea growth.' 4. Say: 'Now complete the CER frame on your own — 5 minutes.' Circulate and read 3–4 CER	Claim–Evidence–Reasoning (CER) Frame — this structured writing strategy (NGSS SEP 6: Constructing Explanations) requires learners to distinguish between data (evidence) and interpretation (reasoning), and to make an explicit logical connection between the two. Mapping graph regions back to the phenomenon grounds abstract graph analysis in the concrete, memorable image of the pale sukuma wiki under the mango tree, reinforcing the storyline thread.	Observable indicators: (1) CER frame — Claim is a full declarative sentence; Evidence includes at least two numerical values from the data table with units; Reasoning explicitly uses the term 'limiting factor' and explains the mechanism. (2) Phenomenon connection — learner can point to or describe a specific region of the graph (low-intensity plateau region) and link it to the pale, slow-growing shaded plant. Red flag: learners who write only 'more light = more photosynthesis' without acknowledging the levelling-off — teacher prompts: 'Look at the right side of your graph — what happens there?'

	Finally, learners explicitly connect back to the phenomenon: 'Use your graph to explain why the sukuma wiki under the mango tree grew slowly and turned pale — cite a specific region of your graph.'	frames. Look for: (a) claim is a statement not a question; (b) evidence has actual numbers; (c) reasoning uses 'limiting factor.' 5. Phenomenon reconnection: 'Point to the region of your graph that represents the sukuma wiki under the mango tree. What is happening there? Now point to the region representing the sukuma wiki in full midday sun. What is happening there — and does your graph tell you anything about WHY it might wilt?' (Plant flag for Lesson 10: high light = high photosynthesis rate, but also high temperature = high transpiration — the graph cannot explain wilting alone.)		
Driving Question Board (DQB) Creation  VIDEO: Dependent & independent variables: graphing Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Analyze relationships between variables <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the class Driving Question Board. Each group receives two sticky notes of different colours. On the ORANGE note, they write one NEW question their investigation raised (e.g., 'If CO₂ is also a limiting factor, why don't farmers in Nairobi pump extra CO₂ into their greenhouses?'; 'Does temperature affect photosynthesis the same way light does?'; 'Why does the sukuma wiki wilt even though water is in the soil — our graph doesn't explain that?'). On the GREEN note, they write one question from the board that today's lesson PARTIALLY OR FULLY answered, and a one-sentence summary of the answer. Groups post both notes on the board. The teacher reads 3–4 orange notes aloud and clusters them under emerging themes: 'RATE AND LIMITING FACTORS', 'TEMPERATURE', 'WATER/WILTING'. The teacher explicitly flags the wilting cluster:	1. Say: 'You have 4 minutes — two sticky notes. Orange: one new question your experiment or graph raised. Green: one question from the board you can now answer, plus your answer in one sentence.' 2. After posting, scan the board and read 4 orange notes aloud, asking the class: 'Is this question related to limiting factors? To wilting? To something else?' Sort them into clusters verbally. 3. Explicitly point to any notes about wilting or water and say: 'This is our unfinished business. Our experiment today measured light intensity — but the sukuma wiki phenomenon includes wilting in wet soil. What process involves water leaving the plant? We will tackle that later in the unit.' 4. Update the board's ANSWERED/PARTIALLY ANSWERED column visibly — use a different colour marker to tick questions addressed today. 5. Say: 'Look at the board. We	Driving Question Board — two-colour sticky note protocol distinguishes between knowledge consolidation (green) and productive uncertainty (orange). Publicly clustering new questions by theme models scientific reasoning about what is known vs. unknown, and creates explicit narrative continuity between lessons. Flagging the wilting gap sustains the overarching phenomenon as the engine of inquiry across the unit.	Observable indicators: (1) Every group posts both a green and an orange note — teacher checks before moving on. (2) Green notes contain a specific answer referencing data or the limiting factor concept, not just 'we learned about photosynthesis.' (3) Orange notes are genuine questions, not statements. Red flag: groups posting vague green notes like 'light is important' — teacher prompts: 'Can you add a number from your data to make that more precise?'

	'Notice — our graph today told us a lot about light intensity and rate of photosynthesis, but it does NOT explain the wilting. That gap is what Lesson 8 is about.'	started with one puzzling observation about a student's kitchen garden in Nairobi. Look how many questions we have now answered — and how many new, deeper questions we have generated. That is what real science looks like.'		
Model Building Phase  VIDEO: Cell size <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Limiting Factors to Population Growth (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners retrieve their cumulative photosynthesis model (begun in earlier lessons). They are asked to add TWO new features to the model based on today's evidence: (1) An ARROW or DIAL labelled 'Light Intensity' that can be shown at different levels — learners annotate that a higher dial setting produces more O₂ and glucose (faster rate), while a low dial setting (shade) produces less — connecting directly to the pale sukuma wiki. (2) A 'RATE CEILING' annotation or a small embedded graph sketch showing that above a certain light intensity, something else (CO₂, temperature) becomes the limiting factor. Groups then conduct a 2-minute Gallery Walk to view two other groups' models. Each group leaves one sticky note with a 'GLOW' (something the model shows clearly) and one 'GROW' (something missing or that could be clearer). Learners return to their model and make one final revision based on the feedback received. The lesson closes with the teacher asking the whole class: 'Our model is getting more and more detailed. What is it still missing to fully explain why sukuma wiki wilts at midday even when the soil is wet?' (Expected answer: the model shows photosynthesis inputs/outputs but does not yet show water leaving the leaf —	1. Say: 'Get out your class model. You are going to add two new pieces of evidence from today.' Read aloud the two tasks from the board. Give groups 8 minutes. 2. Circulate and check that both additions are present. Prompt groups who have not added the rate ceiling: 'What did your graph look like at high light intensity? Did the rate keep going up forever? Show that on your model.' 3. After 8 minutes, say: 'Gallery Walk — 2 minutes. Walk to two other models. Leave a GLOW note and a GROW note. Be specific.' 4. After the gallery walk, give groups 2 minutes to read feedback and make one revision. 5. Whole-class close: Cold-call two learners: 'Looking at our model — what process involving water is still completely missing?' WAIT 12 seconds. If no response: 'Think about what happens to water inside the leaf on a hot day — does it all stay inside?' 6. Final statement: 'In Lesson 8, we will add the missing piece — water movement through the plant and out of the leaves. That will finally let us explain the morning wilting of the sukuma wiki.'	Cumulative Model Revision with Gallery Walk Feedback — iterative model building (NGSS SEP 2: Developing and Using Models) makes thinking visible and tracks conceptual growth across the lesson sequence. The Gallery Walk introduces peer critique as a scientific norm. Deliberately leaving a gap in the model (transpiration) maintains the driving question as an unresolved, generative puzzle, ensuring learners arrive at Lesson 10 with genuine intellectual need to know.	Observable indicators: (1) Every group's model now includes a light-intensity arrow/dial with annotations linking low intensity to slow rate and the pale sukuma wiki. (2) A rate-ceiling notation or embedded graph sketch is visible on the model. (3) GLOW/GROW sticky notes are specific (e.g., 'GROW: your model shows CO₂ going in but doesn't show where water goes'). (4) Exit indicator: at least 2 out of 4 cold-called learners can articulate that transpiration/water loss is the missing piece. Red flag: models that show only the photosynthesis equation without rate or limiting factor notation — teacher stays with that group during model revision.

	transpiration. This seeds Lesson 10.)			
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D. TEACHER REFLECTION

After the lesson, reflect on the following: \n1. INVESTIGATION QUALITY: Did learners successfully identify all three variable types (independent, dependent, controlled) before collecting data? If not, which variable caused the most confusion — and how will you address experimental design more explicitly in future planning phases? \n2. GRAPH INTERPRETATION: Could learners describe the shape of the rate–intensity curve AND explain the levelling-off in terms of limiting factors? If learners described only the rising portion, plan a brief whole-class graph annotation activity at the start of Lesson 8 to address the plateau. \n3. PHENOMENON CONNECTION: When asked to point to the region of the graph representing the shaded sukuma wiki, could learners do so confidently? The quality of this connection reveals whether the storyline is functioning as intended or whether the phenomenon needs to be re-centred more explicitly. \n4. DQB GAP IDENTIFICATION: Did the orange sticky notes on the DQB include questions about wilting and water movement? If not, learners may not yet see the gap between what photosynthesis explains and what transpiration explains — consider opening Lesson 8 with a direct revisit of the wilting puzzle before introducing the new content. \n5. DIFFERENTIATION: Were all groups able to collect or read data and plot a graph, or did some groups spend the full period on the planning phase? If so, consider pre-completing the planning sheet for the weakest groups in Lesson 8 so they can focus on data interpretation and model building.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners counted oxygen bubbles produced by an aquatic plant (or read a simulated dataset) at five different light intensities (lamp distances: 5 cm, 10 cm, 20 cm, 30 cm, 40 cm). They recorded bubble counts per minute in a results table and plotted a line graph of rate of photosynthesis (bubbles min ⁻¹) vs light intensity.
What did I learn?	The rate of photosynthesis increases as light intensity increases, but only up to a maximum — above that point the curve flattens, indicating that another factor (CO ₂ concentration or temperature) has become the limiting factor. A LIMITING FACTOR is the factor present at the lowest level that controls the rate of a process. The pale, slow-growing sukuma wiki under the mango tree is in the low-light-intensity region of the graph where photosynthesis rate is low.
How does this explain the phenomenon?	Why the sukuma wiki under the mango tree grows slowly and turns pale: the shade reduces light intensity, placing the plant in the region of the rate–intensity graph where photosynthesis rate is low. Insufficient glucose is produced for growth, and insufficient energy is available for chlorophyll synthesis, causing pallor. However, the wilting puzzle remains unexplained by this graph — water movement out of the leaf is a missing piece that will be addressed in Lesson 10.

LESSON 8 (80 minutes): Factors Affecting Photosynthesis Rate — Data Analysis & Explain

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To analyse experimental data and apply the concept of limiting factors to explain how CO ₂ concentration and temperature affect the rate of photosynthesis, and to connect these factors to real agricultural decisions in Kenya.
Knowledge	Learners will know that the rate of photosynthesis is controlled by limiting factors (light intensity, CO ₂ concentration, temperature), and that at any moment the factor in shortest supply determines the overall rate.
Skills	Learners will be able to interpret line graphs of photosynthesis rate vs. a single variable, identify limiting factors from data, and construct written explanations linking evidence to the anchoring phenomenon.
Attitudes	Learners appreciate that scientific data analysis — not guesswork — underpins smart farming decisions, and value collaborative reasoning when interpreting ambiguous evidence.
Key Inquiry Question	What factors affect the rate at which plants make food?
Purpose in Storyline	In Lessons 1–7 learners built understanding of the light reactions, the Calvin cycle, and the role of water. Lesson 8 completes the evidence-gathering sequence by zooming out: now that learners know the machinery of photosynthesis, they use data to explain what 'speeds up' or 'slows down' that machinery — directly answering why the sukuma wiki under the mango tree grew pale and slowly (low light → low rate) and setting up the final two lessons on agricultural application and summative assessment.
Safety Notes	No hazardous materials are used in this lesson. All data sets are printed or displayed digitally. Remind learners to handle printed graph sheets carefully and to sit safely when working in groups. If the lesson is conducted outdoors, learners should remain within the designated school compound.

B. LESSON OVERVIEW

Learners open by revisiting the sukuma wiki phenomenon and their Lesson 7 graphs (light intensity vs. photosynthesis rate). They then receive two new printed datasets — one for CO₂ concentration and one for temperature — and work in groups to plot, annotate, and interpret the graphs. A structured gallery walk lets groups critique each other's reasoning. The teacher leads a whole-class Claim–Evidence–Reasoning (CER) discussion to consolidate the concept of limiting factors. Learners apply this concept to a Kenyan agricultural scenario (CO₂ enrichment in Naivasha greenhouse tomato farms) and update both their Driving Question Boards and their cumulative photosynthesis models. The lesson closes with an exit ticket that checks whether learners can identify a limiting factor from a novel data set.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Photosynthesis: Calvin cycle <i>Related topic: not</i>	Learners are shown two blank axes on the board: (A) Rate of photosynthesis vs. CO ₂ concentration, and (B) Rate of photosynthesis vs. Temperature.	Display the two sets of blank axes on the board or projector. Say: 'Before I give you any data, I want your brain's best prediction. Use what you already know about how	Individual prediction sketching followed by partner comparison activates prior knowledge from Lessons 5–7 and creates cognitive dissonance when real data	Observable indicator: Each learner has a sketch with a labelled axis and at least one written justification sentence in their notebook before group work

<i>an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Limiting Factors to Population Growth (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Working individually in their science notebooks for 3 minutes, they sketch what they think each graph will look like and write one sentence justifying each prediction. They then share predictions with a 'turn-and-talk' partner (1 minute) before three pairs are cold-called to share with the class.	photosynthesis works — the light reactions, RuBisCO, stomata — to sketch these two graphs. You have 3 minutes. Work alone and in silence.' Set a visible timer. After 3 minutes, say: 'Now share your sketch with your partner. Do you agree or disagree? You have 1 minute.' After the partner talk, cold-call exactly three pairs (choose one high-confidence, one uncertain, one who has not yet spoken this week). For each, ask: 'Walk us through your thinking — what did you draw and why?' WAIT 12 seconds after each prompt before redirecting. Record the three contrasting predictions on the board under the heading 'Our Predictions — we will test these.' Do NOT correct predictions yet.	diverges from predictions. Recording predictions publicly creates accountability and gives learners a concrete anchor to revise later in the lesson.	begins. Teacher circulates during the 3-minute silent period and notes whether learners are drawing plateau curves, linear curves, or bell curves — this reveals the depth of their mechanistic understanding of enzyme kinetics and stomatal behaviour.
Observe Phase  VIDEO: Photosynthesis: Calvin cycle - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili) Search ARES for similar videos  EXERCISE: Limiting Factors to Population Growth Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Groups of four receive two data-set cards. Dataset A: CO₂ concentration (0.01%, 0.04%, 0.08%, 0.16%, 0.32%) vs. relative rate of photosynthesis (measured in units of O₂ released per minute) at constant light and 25 °C — values show a rapid rise that levels off to a plateau. Dataset B: Temperature (5 °C, 10 °C, 15 °C, 20 °C, 25 °C, 30 °C, 35 °C, 40 °C, 45 °C) vs. relative rate of photosynthesis at constant light and 0.04% CO₂ — values show a rise peaking at ~35 °C then a sharp decline. Groups plot both graphs on provided grid paper (15 minutes), give each graph a title, label both axes with units, and annotate at least two features on each graph with a written label (e.g., 'rate levels off here — plateau', 'sharp drop — enzymes denatured?'). Groups also retrieve their Lesson 7 light-intensity	Distribute data-set cards and grid paper. Say: 'You are scientists receiving new field data from a laboratory in Nairobi. Your job is to turn this data into a graph, then read the story the graph is telling. You have 15 minutes. Every group member must hold a pencil — everyone plots.' Circulate continuously. At the 5-minute mark, check that all groups have chosen appropriate scales — if a group has squashed all points into one corner, ask: 'What would happen to your graph if you spread your x-axis all the way to the edge of the paper? Try it.' At the 10-minute mark, prompt annotation: 'I can see your line — now tell me what's interesting about it. Put a star on the most surprising point and write a label.' At 14 minutes, say: 'Thirty seconds — finish your labels and lay all three graphs side by side on your desk.' Cold-call	Data-to-graph translation (NGSS Science and Engineering Practice 4 — Analysing and Interpreting Data) ensures learners engage directly with the numbers rather than receiving a pre-drawn graph. Placing all three graphs side by side operationalises the concept of multiple variables and primes learners to see a common pattern (plateau behaviour) that points toward limiting factors.	Observable indicator: Each group produces two correctly scaled, labelled, and annotated graphs within 15 minutes. Teacher checks for: (1) appropriate axis scales, (2) smooth curve of best fit rather than dot-to-dot lines, (3) at least two annotations per graph. Groups with only dot-to-dot lines are prompted: 'A scientist draws a line of best fit — what does that tell us that connecting dots does not?'

	graphs and place all three graphs side by side for comparison.	one group to describe what they notice when looking at all three graphs together. WAIT 15 seconds before taking the answer.		
Explain Phase  VIDEO: Breaking down photosynthesis stages <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Scientific method and data analysis <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Step 1 — Gallery Walk (10 minutes): Each group posts their three graphs on the wall or board. Groups rotate around the room with sticky notes and write one 'glow' (something the graph communicates well) and one 'question' (something that puzzles them) on each other's work. Step 2 — Limiting Factor Mini-Lecture with CER (15 minutes): Teacher introduces the term 'limiting factor' using a Kenyan farming analogy, then facilitates a whole-class CER (Claim–Evidence–Reasoning) discussion. Each group constructs a written CER statement for one assigned scenario. Step 3 — Agricultural Application (10 minutes): Groups read a short scenario card about greenhouse tomato farmers in Naivasha who pipe CO₂ into their polytunnels during the day and discuss: 'Using your graphs and the concept of limiting factors, explain why adding CO₂ increases yield — and under what conditions it would NOT help.' Groups share answers; teacher consolidates.	GALLERY WALK: Say: 'You have 10 minutes. Move silently. Read the graphs. Leave one sticky-note glow and one sticky-note question on every group's work except your own. Be specific — write what graph feature you are referring to.' Set timer. Circulate; redirect vague notes ('great graph') by saying: 'Can you point to a specific part of their graph and explain what it tells you?' LIMITING FACTOR LECTURE (5 minutes): Say: 'Here is an analogy. Imagine a githeri recipe — you need maize and beans. If you have plenty of beans but only a little maize, the amount of githeri you can make is limited by the maize. The maize is the limiting factor. Photosynthesis is similar.' Draw a simple three-variable diagram on the board (light, CO₂, temperature each as an arrow feeding into a 'photosynthesis rate' box). Say: 'The factor present in the lowest amount relative to what the plant needs at that moment is the limiting factor.' CER DISCUSSION (10 minutes): Assign each group one of three scenarios: (A) A sukuma wiki plant in a sunny Nairobi garden at 0.04% CO₂ and 30 °C — which factor is limiting? (B) The same plant moved into deep shade — which factor is now limiting? (C) A plant in a heated greenhouse at 45 °C with high CO₂ and full light — what happens to the rate and why? Say: 'Write a CER statement: your Claim (one sentence), your Evidence (quote numbers from your graph), and	The gallery walk deploys Critique and Communication (NGSS SEP 8) so learners evaluate peers' data representations before being given vocabulary. The CER framework (NGSS SEP 6 — Constructing Explanations) scaffolds the transition from 'I see a plateau' to 'CO₂ is limiting because...'. The Naivasha agricultural application transfers the abstract concept to a real Kenyan economic context, deepening relevance and testing flexible understanding.	Observable indicator: Each group's written CER statement includes a specific data value from their graph as evidence and names a mechanism (e.g., 'RuBisCO cannot find enough CO₂ molecules', 'enzymes denature above 40 °C') in the reasoning. Teacher collects CER sheets at the end of Phase 3 and uses a rapid 2-minute scan during Phase 4 to identify misconceptions for the next lesson. Red flag: any group whose reasoning cites only correlation ('as CO₂ goes up, rate goes up') without a mechanistic explanation.

		your Reasoning (explain the mechanism). You have 5 minutes.' Cold-call one group per scenario. WAIT 12 seconds after asking 'Does anyone want to challenge or add to that reasoning?' NAIVASHA APPLICATION (10 minutes): Distribute scenario cards. Say: 'Read this card carefully. Use your CO₂ graph specifically — put your finger on the relevant part of the graph — and explain whether adding CO₂ will always help.' After 5 minutes of group discussion, cold-call two groups. Ask the second group: 'Do you agree with Group 1's reasoning, or do you see something they missed?' WAIT 12 seconds.		
Driving Question Board (DQB) Creation  VIDEO: Photosynthesis: Calvin cycle <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Limiting Factors to Population Growth (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Each learner receives two sticky notes. On the first (green), they write one thing they can now explain about the sukuma wiki phenomenon that they could not explain at the start of the unit. On the second (orange), they write one new question that today's lesson has raised for them. Learners post both notes on the class DQB in the appropriate columns ('What we can now explain' and 'New questions'). The teacher reads three orange questions aloud and the class votes with a show of hands on which is most pressing — this question is circled on the DQB as the class's priority for Lessons 9–10.	Distribute sticky notes. Say: 'Our Driving Question asks why the sukuma wiki wilts in the midday sun and goes pale in the shade. Based on today's data analysis, write on your green note: one part of that phenomenon you can now explain using the concept of limiting factors. Be specific — mention a factor and a mechanism.' Allow 2 minutes of silent writing. Say: 'On your orange note, write the most burning question today's lesson has left you with. It can be about plants, farming, the data, or anything connected.' Allow 90 seconds. 'Come up and post both notes.' After posting, read three orange questions aloud and say: 'Show me with your hand — which of these three questions do you most want answered before the unit ends?' Tally votes visibly. Circle the winning question and say: 'We will make sure Lesson 9 addresses this.'	The two-colour DQB update makes learners' growing understanding visible and metacognitive (they must judge what they now know vs. what remains unknown). The class vote on the priority question models how scientific communities decide what to investigate next, and gives learners genuine agency in the unit's closing direction.	Observable indicator: Green sticky notes mention at least one specific factor (light, CO₂, or temperature) and one mechanism (e.g., 'stomata close at midday reducing CO₂ entry'). Orange sticky notes are genuinely novel — not repetitions of questions already on the DQB. Teacher scans the green notes during Phase 5 to check for persistent misconceptions before the next lesson.

Model Building Phase  VIDEO: Photosynthesis: Calvin cycle - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Limiting Factors to Population Growth Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners retrieve their cumulative photosynthesis model (built across Lessons 3–7) and add a new 'Limiting Factors Layer' using a different colour pen. They annotate the model to show: (1) where in the photosynthesis process each limiting factor acts (e.g., temperature affects enzyme activity in the stroma; CO₂ availability affects the Calvin cycle; light affects the light reactions); (2) an arrow labelled 'rate bottleneck' pointing to the step controlled by whichever factor is most limiting. Pairs compare models and discuss one difference between their annotations. The lesson closes with a 3-2-1 exit ticket.	Say: 'Take out your cumulative model. You are going to add a new layer — use your coloured pen so it is clear what you added today. Draw an annotation for each of the three limiting factors: where does it act in the photosynthesis process, and what happens to the rate when it runs out?' Allow 7 minutes. Circulate and prompt with: 'You have CO₂ written here — can you draw an arrow showing exactly which step in the Calvin cycle is affected?' and 'If temperature rises to 45 °C, which part of your model breaks down first — the chlorophyll, the enzymes, or the membrane?' After 7 minutes, say: 'Compare your model with your partner's. Find one annotation where you drew the limiting factor acting in a different place. Discuss for 2 minutes — who has stronger evidence for their choice?' Cold-call one pair to share their disagreement and resolution. WAIT 12 seconds. EXIT TICKET (3 minutes): Project or write: 'A farmer in Kericho grows tea in cool, misty conditions (low temperature, moderate light, normal CO₂). She adds a fertiliser that boosts CO₂ release in the soil. Will this increase her photosynthesis rate significantly? Use the concept of limiting factors and refer to a specific graph from today to justify your answer.' Learners write individually on a half-sheet and hand it in as they leave.	Model revision (NGSS SEP 2 — Developing and Using Models) makes learning cumulative and visible — learners can physically see their understanding deepen across lessons by looking at the layers of annotation. The exit ticket deliberately places the limiting factor concept in a novel Kenyan context (Kericho tea, cool and misty — temperature is likely limiting, not CO₂) to assess transfer rather than recall.	Observable indicator on model: At least three new annotations are added in the new colour, each correctly linking a limiting factor to a specific step in photosynthesis (light reactions or Calvin cycle). Exit ticket rubric: Full credit — correctly identifies temperature (not CO₂) as the limiting factor in cool Kericho conditions, references the temperature graph plateau or decline, and explains why adding CO₂ alone will not help when temperature is the bottleneck. Partial credit — identifies a limiting factor but does not reference graph data. No credit — states CO₂ addition will simply increase rate without considering temperature. Teacher uses exit ticket responses to group learners for targeted support in Lesson 9.
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D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Were learners able to distinguish between the plateau region (where a different factor is now limiting) and the rising region of the CO₂ and temperature graphs? If many groups annotated both graphs identically without recognising the bell-curve shape of the temperature graph, plan a

brief 5-minute re-teach at the start of Lesson 9 using the enzyme denaturation concept. (2) Did the Naivasha greenhouse scenario generate genuine transfer, or did learners simply repeat the textbook definition of limiting factor? If learners struggled to specify under which conditions adding CO₂ would NOT help, consider adding a second scenario card in Lesson 9 that forces them to reason about two variables changing simultaneously. (3) Review the exit ticket responses before Lesson 9: sort them into three piles (full credit, partial, no credit). Use the no-credit pile to identify the most common misconception — likely that 'more of anything always helps'. Plan a targeted probe question for the opening of Lesson 9 to address this. (4) Did the DQB update surface any questions about topics not yet covered (e.g., mineral salts, water potential, respiration at night)? If so, note these for the unit summary in Lesson 10. (5) Reflect on group equity: did the same learners dominate the CER discussion? If yes, assign specific roles (Evidence Reader, Claim Writer, Reasoning Challenger) for the group tasks in Lesson 9 to ensure broader participation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners plotted and annotated line graphs showing how CO ₂ concentration and temperature each affect the rate of photosynthesis, and compared these with their Lesson 7 light-intensity graphs.
What did I learn?	The rate of photosynthesis is controlled by limiting factors — light intensity, CO ₂ concentration, and temperature — and the factor in shortest supply at any moment determines the overall rate. Temperature graphs show a bell-curve shape because enzymes denature above ~40 °C.
How does this explain the phenomenon?	Learners used the limiting factor concept to explain why the sukuma wiki under the mango tree grew pale and slowly (low light was the limiting factor), why greenhouse tomato farmers in Naivasha add CO ₂ to boost yields (CO ₂ was the limiting factor in well-lit, warm greenhouses), and why adding CO ₂ to cool, misty Kericho tea farms would not significantly increase the photosynthesis rate (temperature, not CO ₂ , is limiting there).

LESSON 9 (80 minutes): Mineral Nutrition: What Does the Soil Give the Plant?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to identify essential mineral nutrients required by plants, describe their roles in plant growth and metabolism, and match observable deficiency symptoms to specific mineral deficiencies using KALRO secondary data and photographic evidence.
Knowledge	Learners will identify macronutrients (nitrogen, phosphorus, potassium, calcium, magnesium, sulfur) and micronutrients (iron, manganese, zinc, copper, boron, molybdenum), state the specific physiological roles of each mineral in plant growth, and describe the deficiency symptoms associated with key minerals including chlorosis, necrosis, stunted growth, and purpling of leaves.
Skills	Learners will analyse secondary data tables and photographs from KALRO field research to match deficiency symptoms to causal minerals, construct a structured comparison table of macro- and micronutrients, and draw evidence-based conclusions that connect mineral uptake through roots to the overall health and productivity of the sukuma wiki plant.
Attitudes	Learners will appreciate the scientific basis of soil fertility management and develop curiosity about how invisible chemical elements dissolved in soil water determine whether a plant thrives or struggles, linking this to the livelihoods of Kenyan smallholder farmers.
Key Inquiry Question	How do plants make their own food? What factors affect the rate at which plants make food?
Purpose in Storyline	In Lessons 1–8 learners established that sukuma wiki needs light for photosynthesis, water for turgor and as a raw material, CO ₂ as a carbon source, and chlorophyll as the light-capturing pigment. Yet photosynthesis alone cannot explain all plant growth — the plant also needs mineral elements for building proteins, nucleic acids, chlorophyll, and cell walls. Lesson 9 closes this gap by introducing mineral nutrition as the final major input into plant nutrition, explaining why a Nakuru maize farmer whose plants show yellowing older leaves is experiencing nitrogen deficiency, and why our sukuma wiki in the shade under the mango tree gradually turned pale. Learners now hold a nearly complete picture of what the plant needs to stay green, grow, and make its own food.
Safety Notes	No hazardous chemicals are used in this lesson. Learners handle printed photographs and data tables only. Remind learners that real fertiliser compounds (especially nitrogenous fertilisers) must be stored and handled carefully by adults; they are not to be tasted or handled without protective gloves in a practical setting. Wash hands after handling any soil samples if brought into class.

B. LESSON OVERVIEW

Learners open by revisiting the pale, slow-growing sukuma wiki under the mango tree and the yellowing Nakuru maize field (a supporting phenomenon introduced earlier in the sequence) to generate predictions about what might be missing beyond light. They then analyse a curated set of KALRO crop-deficiency photographs and a secondary data table showing mineral functions and deficiency symptoms across common Kenyan crops. Through a structured Claim–Evidence–Reasoning (CER) activity, each group matches a deficiency photograph to a specific mineral and justifies their reasoning. The class builds a shared Mineral Nutrition Reference Table. Learners then revisit the driving question, updating their Driving Question Board (DQB) and revising their cumulative plant nutrition models to include mineral uptake via root hair cells as an additional input arrow alongside water, CO₂, and light.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: plant nutrition mineral macronutrients  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: plant nutrition mineral macronutrients	Learners are shown two side-by-side images on the ARES platform: (A) healthy, deep-green sukuma wiki growing in a well-managed Kiambu kitchen garden, and (B) the same variety of sukuma wiki growing under a mango tree — pale yellowish-green leaves, thin stems, slow growth — as established in the phenomenon. A third image (C) shows a maize field in Nakuru where older lower leaves are turning uniformly yellow while younger upper leaves remain green. Learners individually write a 3-sentence prediction in their exercise books answering: 'Beyond light, what else might be missing or insufficient in images B and C? What do you think the soil provides to a plant?' They then share predictions with a shoulder partner before three pairs are cold-called to share with the class. Teacher records all distinct prediction categories on the board under the heading 'What could be missing?'	Display images A, B, and C and say: 'We have spent eight lessons building our understanding of what sukuma wiki needs. Light — yes. Water — yes. Carbon dioxide — yes. Chlorophyll — yes. But look carefully at image B and image C. These plants have water in the soil. They are alive. Yet something is clearly wrong. In your books, write three sentences: what do you think is missing, and what do you think soil actually provides to a plant beyond water?' Allow 3 minutes of independent silent writing. After 3 minutes say: 'Share your prediction with your shoulder partner — 90 seconds.' Then: 'I am going to cold-call three pairs. Pair one — [name student] — what did your pair predict?' After each response, probe with: 'What evidence from the image made you think that?' [WAIT TIME: 12 seconds after each probe.] Cold-call a total of 3 pairs. Record all unique ideas on the board. Do NOT confirm or deny any prediction — say: 'These are all interesting ideas. Today's evidence will help us decide which predictions are correct.'	Prediction Anchor: By committing predictions to writing before seeing data, learners activate prior knowledge about plant inputs (from Lessons 1–8) and create a cognitive hook. The teacher's neutral recording of all ideas without evaluation maintains intellectual safety and ensures learners are genuinely motivated to test their ideas against the KALRO evidence.	Observable indicator: Every learner has written at least three sentences in their book before discussion begins. Teacher circulates during the 3-minute writing period and notes (without correcting) whether learners are connecting their predictions to the phenomenon (mango shade, wet soil, yellowing) or making generic statements. If fewer than half the class mentions 'soil' or 'nutrients' unprompted, the teacher makes a brief bridging remark before Phase 2: 'Interesting — several of you mentioned the soil. Let us now look at some scientific data about what soil actually contains that plants need.'
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: plant nutrition mineral macronutrients  READING: No closely	Learners work in groups of four. Each group receives: (1) a printed A3 KALRO Secondary Data Table showing eight minerals (N, P, K, Ca, Mg, S, Fe, Zn), their chemical symbols, classification as macro or micro, their role in the plant, and the deficiency symptom observed in Kenyan crops; (2) a set of six laminated deficiency photographs (or ARES platform image gallery) labelled only with letters A–F showing: nitrogen-deficient maize from Nakuru (yellowing of older leaves), phosphorus-deficient	Distribute materials and say: 'Your task is to become plant diagnosticians — like the scientists at KALRO in Nairobi who advise Kenyan farmers on why their crops are failing. For each photograph, use the data table to make a Claim about which mineral is deficient, cite the Evidence you can see in the photograph, and write your Reasoning — connecting what you see to what the data table says the deficiency looks like.' Circulate continuously. At the 6-minute mark, check in with one group and	Claim–Evidence–Reasoning (CER) Framework: This strategy requires learners to move beyond pattern-matching to explicit scientific argumentation. By naming a claim, citing specific visual evidence, and linking it to mechanistic data in the table, learners practise the NGSS Science and Engineering Practice of Constructing Explanations. The CER structure also makes learner thinking visible and auditable for formative assessment.	Observable indicator: Each group's CER worksheet shows six completed rows with all three columns filled. Teacher looks specifically for quality of reasoning — does the learner cite the specific symptom (e.g., 'yellowing starts on older leaves') and link it explicitly to the data table (e.g., 'the table says nitrogen deficiency causes chlorosis in older leaves first because nitrogen is remobilised from old to young leaves')? Groups that write only 'it looks yellow' without mechanistic

<i>matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: plant nutrition mineral macronutrients	bean plants from Kisii (purple-tinged leaves and stems), potassium-deficient sukuma wiki (brown leaf margins/scorching), magnesium-deficient tea from Kericho (interveinal chlorosis on older leaves), iron-deficient spinach from Nairobi urban farm (interveinal chlorosis on young leaves), and zinc-deficient maize from the Rift Valley (white striping on leaves). Groups study the photographs and data table and complete a structured CER worksheet: for each photograph, write a Claim (which mineral is deficient), Evidence (specific visual symptom observed), and Reasoning (how the data table supports the match). Groups have 18 minutes for this activity.	ask: 'You have matched photograph C to potassium — walk me through your reasoning.' [WAIT TIME: 10 seconds.] At the 12-minute mark, identify a group that has made an error (e.g., confusing magnesium and iron interveinal chlorosis) and ask: 'Both of these show interveinal chlorosis. How does the data table help you tell them apart?' [WAIT TIME: 15 seconds.] Key distinction to guide toward: magnesium deficiency appears on OLDER leaves first (magnesium is mobile in the phloem); iron deficiency appears on YOUNGER leaves first (iron is immobile). Do not give the answer — ask: 'Which leaves are affected — old or young? Now check the table again.' At 18 minutes, call time.		reasoning are prompted: 'Why older leaves and not younger ones? What does the table tell you about whether this mineral moves around inside the plant?'
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: plant nutrition mineral macronutrients  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: plant nutrition mineral macronutrients	Each group shares one CER match with the class in a gallery-style debrief (2 minutes per group, 6 groups × 1 photograph each = 12 minutes). As groups present, the class co-constructs a shared Mineral Nutrition Reference Table on the board with columns: Mineral Symbol Macro/Micro Role in Plant Deficiency Symptom Kenyan Crop Example. After all six photographs are debriefed, learners individually write a 'So What?' paragraph in their books connecting mineral nutrition to the sukuma wiki phenomenon: 'Explain why the sukuma wiki under the mango tree became pale and grew slowly even though the soil was wet.' They must use at least three mineral names and the terms 'macronutrient', 'chlorophyll', and 'root uptake' in their paragraph. Finally, the class discusses how root hair cells	Facilitate the gallery debrief by assigning each group a photograph. Say: 'Group 1 — you have 2 minutes to present your CER for photograph A. Tell us your claim, your evidence, and your reasoning. The rest of the class: listen and add the information to your own copy of the reference table.' After each presentation, ask the class: 'Does anyone want to challenge or add to their reasoning?' [WAIT TIME: 12 seconds.] Cold-call at least one non-presenting learner per photograph with: '[Name], do you agree with their claim? Why or why not?' After all six presentations, say: 'Now I want you to write independently for 5 minutes. Connect what you have just learned to our sukuma wiki story. Why did it turn pale under the mango tree? You must use the words macronutrient, chlorophyll,	Gallery Debrief with Co-constructed Reference Table: Distributing the explanatory work across groups means every learner hears six worked examples of CER reasoning, while the act of co-constructing the table together gives the class a shared artefact that encodes the lesson's core content. The 'So What?' paragraph forces individual sense-making and is a direct rehearsal for examination-style extended-answer questions. Connecting active transport back to cell biology (Strand 1) reinforces vertical coherence across the curriculum.	Observable indicator: The 'So What?' paragraph uses all three required terms correctly in context. Teacher collects or photographs 4–5 paragraphs from different ability levels during the lesson to review after class. A strong paragraph reads approximately: 'The sukuma wiki under the mango tree received less light, so it photosynthesised less. But even if light were sufficient, the plant also needs macronutrients like nitrogen to build chlorophyll, which is why leaves turned pale — this is chlorosis caused by nitrogen deficiency. Minerals are absorbed by root uptake through root hair cells by active transport. Without enough nitrogen, the plant cannot make chlorophyll or proteins, so it grows slowly.' Any paragraph that conflates 'minerals' with 'food' (i.e., says the plant 'eats minerals') is flagged for correction — minerals

	absorb mineral ions by active transport (linking to prior knowledge of cell biology), and why mineral deficiency in the sukuma wiki is a plant nutrition problem — not just a soil problem.	and root uptake.' After 5 minutes, cold-call 2 learners to read their paragraphs aloud. Then ask the whole class: 'We said in Lesson 3 that root hair cells absorb water by osmosis. How do you think mineral ions get into the root — is it the same process?' [WAIT TIME: 15 seconds.] Guide learners to recall that ions move by active transport (requiring energy from respiration), not osmosis, linking mineral uptake to the broader energy story of the plant.		are not the plant's food; glucose is. This is a critical misconception to address before Lesson 10.
Driving Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: plant nutrition mineral macronutrients  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: plant nutrition mineral macronutrients	Learners return to the class Driving Question Board. Working individually, each learner writes on a sticky note one question that Lesson 9 has answered and one new question it has raised, then places them in the appropriate columns ('Answered' and 'Still Wondering'). The class reviews the board together for 5 minutes. The teacher highlights how many of the original DQB questions (posted in Lesson 1) about 'what does soil give the plant?' and 'why are leaves green?' can now be answered with precision. Learners also note which questions remain open and will be addressed in the final Lesson 10 synthesis.	Hand out two sticky notes per learner. Say: 'Look at our Driving Question Board. Think about what you personally now understand that you did not understand before this lesson. Write one thing Lesson 9 has answered — be specific, use the mineral names. Then write one question this lesson has raised for you or that you still cannot answer.' Allow 3 minutes of silent writing. Say: 'Place your answered note in the green column and your still-wondering note in the orange column.' After all notes are placed, read 3–4 'still wondering' notes aloud and say: 'These are exactly the kinds of questions scientists ask. Notice that [student's name] is asking whether a plant can recover from nitrogen deficiency if you add fertiliser late — that is a real question Kenyan agronomists at KALRO study.' Point to the original driving question displayed at the top of the board and ask: 'How much of this question can we now answer? What piece is still missing?' [WAIT TIME: 12 seconds.] Expected response: learners can now explain the greenness (chlorophyll, needs Mg and N), the growth (needs N, P,	DQB Progress Audit: Asking learners to explicitly distinguish 'answered' from 'still wondering' questions trains metacognitive monitoring — a key competency in CBE. The public nature of the board means learners see their collective intellectual progress across the nine-lesson sequence, which builds motivation and a sense of epistemic momentum toward the final synthesis lesson.	Observable indicator: The 'answered' sticky note names at least one specific mineral and its role (e.g., 'nitrogen is needed to make chlorophyll, so without it leaves go yellow'). A vague note saying only 'I learned about minerals' signals shallow encoding and that learner should be targeted for additional support in Lesson 10. The teacher photographs the board at the end of the phase to track the distribution of remaining questions and use them to plan the Lesson 10 synthesis.

		K), but the wilting mechanism (stomata, transpiration rate) was addressed in earlier lessons — Lesson 10 will synthesise everything.		
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: plant nutrition mineral macronutrients  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: plant nutrition mineral macronutrients	Learners retrieve their cumulative plant nutrition models begun in Lesson 1 and progressively revised through Lessons 2–8. Each learner adds a new layer to their model: (1) a soil zone below the root showing dissolved mineral ions (labelled N, P, K, Ca, Mg, S) in the soil water; (2) an arrow from soil through root hair cells into the plant labelled 'active transport of mineral ions (energy from respiration required)'; (3) annotation arrows from the mineral ions to the plant structures they support — N and Mg to the chloroplast/chlorophyll, P to the cell membrane and DNA, K to stomatal guard cells, Ca to the cell wall; (4) a revised annotation on the chloroplast noting that 'chlorophyll synthesis requires N and Mg — deficiency → chlorosis → reduced photosynthesis → pale leaves and slow growth.' Learners then write a one-sentence model caption: 'This model shows that a sukuma wiki plant needs light, water, CO₂, AND dissolved mineral ions from the soil to photosynthesise, grow, and stay healthy.' Pairs swap models and give one piece of specific feedback using the sentence frame: 'Your model clearly shows _____. You could strengthen it by adding _____.'	Say: 'Take out your plant nutrition models. You are about to make them much more powerful. By the end of today, your model must show not just where energy comes from, but what the soil contributes — and exactly where in the plant each mineral goes.' Display a partially completed annotated model on the board (ARES platform projection) that shows the soil zone but leaves all mineral labels and destination arrows blank. Say: 'Do not copy mine. Yours will be better than mine because you have just spent this lesson gathering evidence.' Allow 10 minutes for individual model revision. Circulate and look for: (a) learners who draw mineral ions entering via the stem or leaves rather than roots — correct by asking 'Where does the plant absorb water from? Minerals are dissolved in that same soil water — so where must they enter?'; (b) learners who connect all minerals only to the leaf — prompt with 'Where does phosphorus go? Check your reference table.' After 10 minutes, say: 'Swap with your partner. Give them one specific compliment and one specific suggestion using the sentence frame on the board. You have 2 minutes.' After peer feedback, ask two volunteers to display their models using the document camera or ARES upload and explain one annotation they added today.	Cumulative Model Revision with Peer Feedback: The cumulative model is the most important epistemic artefact of the ten-lesson sequence. Each revision makes the model more mechanistic and more complete. By requiring learners to annotate with specific destination arrows (mineral → structure → function), the model-building phase demands that learners integrate today's content with all prior lessons. Peer feedback using a structured sentence frame ensures the feedback is specific and constructive rather than generic.	Observable indicator: The revised model shows at minimum (1) dissolved mineral ions in the soil zone, (2) an active transport arrow at the root, (3) at least three correctly labelled destination annotations (e.g., N → chlorophyll, K → guard cells, P → DNA/membrane). A model that shows mineral ions entering through leaves instead of roots, or that fails to connect any mineral to a specific plant structure, requires teacher follow-up before Lesson 10. The one-sentence model caption is collected as an exit artefact — it should name all four major inputs (light, water, CO₂, minerals) to confirm that learners hold an integrated view of plant nutrition.

D. TEACHER REFLECTION

After the lesson, ask yourself the following questions to evaluate the quality of learning and prepare for the final synthesis lesson: (1) During the CER activity, were learners distinguishing between mobile and immobile minerals (i.e., did they use the 'old leaf vs. young leaf' diagnostic correctly)? If not, build a brief 5-minute review into the Lesson 10 opener. (2) Did the 'So What?' paragraphs correctly identify chlorosis as a nitrogen/magnesium deficiency symptom linked to reduced chlorophyll synthesis — or did learners conflate wilting (water/stomata issue from earlier lessons) with chlorosis (mineral deficiency)? Conflation of these two phenomena is the most common misconception in this sub-strand and must be explicitly resolved in the Lesson 10 synthesis. (3) Were learners able to connect active transport of minerals to the energy cost for the plant, linking back to respiration (Sub-Strand 2.3)? If this link was weak, note it as a preview to strengthen when respiration is taught. (4) Look at the DQB 'still wondering' notes — are any new, substantive questions present that deserve airtime in Lesson 10? Incorporate at least one student-generated question into the Lesson 10 driving question revisit. (5) Review the 4–5 collected paragraphs tonight: any learner who says 'the plant eats minerals' or 'minerals are plant food' holds a critical misconception that will undermine their understanding of the distinction between mineral nutrition and carbon nutrition — address this directly at the start of Lesson 10.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners analysed KALRO deficiency photographs and secondary data tables showing mineral deficiency symptoms in Kenyan crops including maize from Nakuru, tea from Kericho, beans from Kisii, and sukuma wiki from Nairobi.
What did I learn?	Plants require macronutrients (N, P, K, Ca, Mg, S) and micronutrients (Fe, Zn, etc.) absorbed from soil water through root hair cells by active transport; each mineral has a specific role in plant structure and metabolism, and deficiency produces characteristic visible symptoms such as chlorosis, necrosis, stunted growth, and purpling.
How does this explain the phenomenon?	The pale yellowing of the sukuma wiki under the mango tree is partly explained by reduced photosynthesis due to low light, but is also consistent with nitrogen and magnesium deficiency causing chlorosis — insufficient mineral uptake means the plant cannot synthesise adequate chlorophyll, proteins, or new cell structures, so it grows slowly and its leaves turn pale even when soil water is available.

LESSON 10 (40 minutes): Model Building, Synthesis and DQB Resolution: Putting the Sukuma Wiki Story Together

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate all learning from the sub-strand by constructing an integrated annotated model that fully explains why sukuma wiki wilts at midday and turns pale in shade, and to connect plant nutrition science to Kenyan food security.
Knowledge	Learners will demonstrate integrated understanding of: leaf structure and its role in photosynthesis and water movement; the two stages of photosynthesis (light-dependent and light-independent) and the molecules they produce; the four raw materials for photosynthesis; limiting factors (light intensity, CO ₂ concentration, temperature) and their effect on photosynthesis rate; macronutrient and micronutrient roles and deficiency symptoms; and the link between plant nutrition and agricultural productivity in Kenya.
Skills	Learners will synthesise information across lessons into a coherent annotated scientific model; construct cause-and-effect explanatory chains; communicate scientific reasoning orally and in writing; evaluate peer models against evidence; and write a structured evidence-based explanation using disciplinary vocabulary.
Attitudes	Learners will appreciate the complexity and elegance of plant biology; value collaborative knowledge-building; show confidence in using scientific evidence to explain real-world phenomena; and recognise the relevance of plant nutrition science to food security and sustainable agriculture in Kenya.
Key Inquiry Question	How do plants make their own food, and what factors determine how well they can do so in real conditions such as a Nairobi kitchen garden?
Purpose in Storyline	This is the final lesson of the sub-strand. Learners return to the anchoring phenomenon — the wilting and paling sukuma wiki — with a full toolkit of evidence gathered over nine lessons. They do not receive new content; instead they integrate leaf structure, photosynthesis stages, limiting factors, and mineral nutrition into one explanatory model, resolve all remaining questions on the Driving Question Board, and produce a written explanation that demonstrates mastery of the driving question.
Safety Notes	No hazardous materials are used in this lesson. If learners are creating poster models using scissors and adhesive, remind them to handle scissors safely. Ensure adequate ventilation if marker pens are used on large paper. No additional safety precautions are required.

B. LESSON OVERVIEW

Lesson 10 is the capstone consolidation lesson for Sub-strand 2.1. The lesson opens with a rapid Predict phase in which learners revisit their original Lesson 1 predictions about the wilting sukuma wiki and note how their thinking has changed. The Observe phase uses a structured gallery walk of anchor images and data compiled from all nine previous lessons, prompting learners to identify the evidence each lesson contributed to the overall explanation. In the Explain phase, small groups construct an integrated annotated model — poster or digital — that links leaf anatomy, light-dependent reactions, light-independent reactions, limiting factors, and mineral nutrition to explain both phenomena (wilting and pallor). Groups present their models and the class uses peer feedback criteria to evaluate them. The Driving Question Board is then cleared and resolved: every remaining question is answered with evidence from the sub-strand. The lesson closes with a three-part individual written synthesis — connecting the sukuma wiki phenomenon to Kenyan agriculture and food security — which serves as the summative performance task for the sub-strand.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Breaking down photosynthesis stages Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Photosynthesis And Transpiration Source: iicba  Search ARES for similar readings	Learners open their exercise books to the prediction they wrote in Lesson 1 about why the sukuma wiki wilts and turns pale. Working individually for 3 minutes, they annotate their original prediction using a different colour pen, crossing out ideas they now know are incorrect and adding brief notes about what they now understand instead. They then share one changed idea with a shoulder partner. Two or three volunteers share with the class. The teacher writes on the board two columns: THEN WE THOUGHT and NOW WE KNOW, and begins populating them from learner contributions.	Display the original anchoring phenomenon photograph or description on the board. Say: 'Go back to your very first prediction from Lesson 1. Read it carefully. Using a red pen or pencil, annotate it — what were you right about? What did you not yet understand? You have three minutes working in silence.' Set a timer. After 3 minutes say: 'Turn to your partner. Tell them one idea you held in Lesson 1 that the evidence from our lessons has changed. You have 90 seconds.' After sharing, cold-call three pairs using name cards. As learners respond, populate the THEN and NOW columns on the board. Do not correct or affirm yet — say: 'We will test these NOW ideas against our models in a few minutes.' WAIT TIME: After asking each learner to share, wait a full 5 seconds before accepting the response, modelling that reflection takes time.	Elicitation of prior reasoning and metacognitive reflection — learners externalise conceptual change, making their learning trajectory visible to themselves and peers. The THEN / NOW board creates a shared record of intellectual growth that the class can reference during model building.	Teacher scans annotated predictions during the 3-minute silent phase, noting which learners still hold misconceptions (e.g., attributing wilting solely to lack of soil water, or failing to link pallor to reduced photosynthesis). These learners are strategically placed in mixed groups for the model-building phase. The THEN / NOW board reveals class-wide gaps that the Explain phase must address.
Observe Phase  VIDEO: Breaking down photosynthesis stages Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Photosynthesis Reactions (Flexbook) Source: CK-12  Search ARES for similar readings	Five stations are set up around the room, each displaying a single A3 summary card compiled from lessons 2 to 9. Station 1: an annotated leaf cross-section diagram (Lesson 2). Station 2: the word and chemical equations for photosynthesis and a list of raw materials (Lesson 3). Station 3: a photograph of the iodine-stained variegated leaf from the starch test practical (Lesson 4). Station 4: a simplified two-stage flow diagram of photosynthesis — light-dependent and light-independent stages — with key molecules labelled (Lessons 5 and 6). Station 5: a graph of photosynthesis rate	Before the gallery walk say: 'Each station holds evidence we have gathered over nine lessons. Your job is not to re-learn it — you already know it. Your job is to harvest it: ask yourself, how does this piece of evidence help explain why the sukuma wiki wilts or turns pale? Fill one row of your Evidence Harvest table at each station.' During the gallery walk, circulate and pose probing questions at each station rather than explaining: At Station 1 ask: 'Which cell layer loses water first when transpiration outpaces absorption — and where do stomata sit?' At Station 3 ask:	Structured gallery walk with an Evidence Harvest table — learners actively retrieve and connect distributed knowledge rather than passively reviewing. The two-column analytical frame (wilting and pallor) keeps attention anchored to the driving question phenomenon throughout observation.	Teacher reviews Evidence Harvest tables during and immediately after the gallery walk to check whether learners are making causal links (e.g., high light intensity increases transpiration rate, stomata open, water loss exceeds uptake, turgor falls, wilting occurs) rather than listing facts in isolation. Persistent surface-level responses signal the need for guided questioning during the Explain phase.

	against light intensity and a mineral deficiency symptom table (Lessons 7, 8, and 9). Groups of four spend 2 minutes at each station. At each station they complete one row of a Evidence Harvest table in their books with three columns: WHAT IT SHOWS, HOW IT LINKS TO WILTING, HOW IT LINKS TO PALLOR. The teacher rings a small bell or claps twice to signal rotation. Total time: approximately 10 minutes including transitions.	'What does the uncoloured section of the leaf tell us about what chlorophyll does?' At Station 5 ask: 'If a plant is nitrogen-deficient, how does that affect its chlorophyll and therefore its photosynthesis rate?' WAIT TIME: Pose each station question, step back, and wait 8 seconds before offering any scaffold — this signals that the question is genuinely open and expects learner reasoning.		
Explain Phase  VIDEO: Photosynthesis: Calvin cycle <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Photosynthesis And Transpiration Source: iicba  Search ARES for similar readings	Groups of four receive one sheet of A2 paper (or access to a shared digital whiteboard) and the following brief: 'Build an annotated model that explains BOTH observations: (1) Why does sukuma wiki wilt at midday when the soil is wet? (2) Why does it turn pale and grow slowly in shade? Your model must include at minimum: a labelled diagram of the leaf cross-section showing stomata and vascular bundles; a summary of the light-dependent and light-independent reactions showing inputs and outputs; arrows showing how limiting factors (light, CO2, temperature) change the rate; and a section on at least TWO minerals and what happens when they are deficient. You have 12 minutes to build your model. Every member of the group must be able to explain any part of it.' After 12 minutes, groups conduct a 90-second standing presentation of their model to the adjacent group (a paired gallery format), then groups swap and repeat. After both presentations, each group receives one written sticky note of peer feedback using the sentence frame: 'Your model	Distribute materials and read the brief aloud. Say: 'Your model is not a beautiful drawing — it is a scientific argument. Every arrow, every label, every word should be doing explanatory work. If you write a word and cannot say what it explains, remove it.' During the 12 minutes of model building, rotate among groups and use targeted prompts: If a group omits the stomata-transpiration link say: 'Your model explains pallor well. But walk me through exactly how the water gets from the wet soil to — and then out of — the leaf on a hot midday. Where does the balance tip?' If a group omits minerals say: 'Your model shows what light does. What does the soil contribute that light cannot replace? Think about what nitrogen and magnesium do inside the chloroplast.' If a group makes an error (e.g., stating glucose is made in the thylakoid) say: 'Interesting — where exactly does CO2 fixation happen, and what molecules from the light stage make it possible?' During presentations, stand at the back and listen. After peer feedback, say: 'You have two minutes. Use	Integrated model construction — the highest-order synthesis task in the lesson. The paired gallery presentation creates authentic audience and accountability. Peer feedback using the structured sentence frame trains learners to evaluate scientific models against criteria (completeness, causal accuracy, explanatory power) rather than aesthetics. Revision in response to feedback models the iterative nature of scientific model-building.	Teacher uses a rapid 4-point rubric (held mentally or on a clipboard) during rotation: 1 = model lists facts without causal links; 2 = model links some elements to wilting OR pallor but not both; 3 = model explains both phenomena with correct causal chains and includes limiting factors and minerals; 4 = model integrates all elements into a single coherent explanatory narrative with accurate science. Groups scoring 1 or 2 receive an additional targeted prompt before presentations. Peer feedback sticky notes are collected at the end of the lesson as secondary formative evidence.

	clearly shows... One thing that would make it stronger is...' Groups have 2 minutes to revise their model in response to the feedback.	the feedback. Make your model stronger.' WAIT TIME: When a group is stuck during model building, ask a question and then physically move away for 10 seconds to prevent immediate rescue — returning only if the group is genuinely unable to proceed.		
Driving Question Board (DQB) Creation  VIDEO: Photosynthesis: Calvin cycle - dubbed(KY) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (Kiswahili)  Search ARES for similar videos  HTML: Photosynthesis Reactions (Flexbook) Source: CK-12  Search ARES for similar readings	The class reassembles facing the Driving Question Board, which has been displayed since Lesson 1. The teacher reads each remaining question card aloud. For each question, a volunteer — or the group whose model best addresses it — provides a one-sentence evidence-based answer. The class votes with thumbs up (fully answered), thumbs sideways (partially answered), or thumbs down (not yet answered) for each response. Questions that receive partial or thumbs-down votes are briefly discussed until a satisfactory answer is reached. Each answered question card is turned over or moved to a RESOLVED column. The driving question itself — Why does sukuma wiki wilt in the midday sun even when the soil is wet, and what does a plant actually need to stay green, grow, and make its own food? — is addressed last, answered collectively by the class in a think-pair-share before the teacher synthesises the class answer on the board.	Stand beside the DQB and say: 'This board has carried our questions since Lesson 1. Today we close it — but only if we can answer every question with evidence. I am going to read each question. If your group model answers it, stand up and give us one sentence — one sentence with evidence, not just a word.' Read each question card slowly. After each volunteer response, facilitate the thumb vote: 'Thumbs: is this question fully answered by science? Up, sideways, or down.' For questions that receive sideways or down votes, say: 'What is still missing from that answer? What evidence from our lessons would strengthen it?' For the driving question itself, say: 'This is the question we started with nine lessons ago. Before I say anything, I want you to think for 30 seconds — what is your best, fullest, most evidence-rich answer to this question right now?' WAIT TIME: Allow 30 full seconds of silence for the driving question think time. Then say: 'Now discuss with your partner for one minute.' Cold-call two pairs, then synthesise on the board. After writing the class synthesis answer, say: 'Amina noticed something in her kitchen garden. Nine lessons later, we can explain it completely. That is what science does.' WAIT	Public DQB resolution — learners experience genuine intellectual closure by returning to questions they themselves generated. The thumb-vote protocol distributes epistemic authority to the class rather than the teacher. Addressing the driving question last, after all sub-questions are resolved, creates a culminating sense-making moment that is earned rather than given.	The quality of one-sentence evidence-based answers reveals whether learners can retrieve and apply integrated knowledge under mild social pressure. The teacher notes which question types remain difficult (e.g., questions about the Calvin cycle intermediates or specific mineral-enzyme relationships) and records these as topics for future review or homework. The thumb-vote distribution across the class provides a quick whole-class comprehension check.

		TIME: After the synthesis, pause for 5 seconds of silence before moving on — to allow the moment of intellectual closure to register.		
Model Building Phase  VIDEO: Photosynthesis: Overview of the light-dependent reactions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: The light-dependent reactions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners now work individually for the final 8 minutes to write a structured three-part evidence-based explanation in their exercise books. The prompt is displayed on the board: PART 1 — THE MIDDAY WILT: Explain in 3 to 4 sentences why sukuma wiki wilts at midday even when the soil is wet. Use the words: stomata, transpiration, turgor pressure, limiting factor, and light intensity. PART 2 — THE SHADE PALLOR: Explain in 3 to 4 sentences why the plant turns pale and grows slowly under the mango tree. Use the words: chlorophyll, light-dependent reactions, ATP, glucose, and nitrogen. PART 3 — THE KENYAN CONTEXT: In 2 to 3 sentences, explain how understanding plant nutrition could help a smallholder farmer in Kiambu County grow healthier sukuma wiki and contribute to food security. You may use ideas from mineral nutrition, limiting factors, or photosynthesis rate. This written explanation is the sub-strand summative performance task. Learners hand it in before leaving or photograph it if digital portfolios are used.	Distribute or display the three-part prompt. Say: 'This is your individual explanation. Your model was your group thinking out loud. This is your thinking on paper — your evidence that you understand the story of plant nutrition from leaf to soil to food on a Kenyan table. Write in sentences. Use the vocabulary words. You have 8 minutes.' During the 8 minutes, circulate silently and read over shoulders. Do not intervene unless a learner is completely unable to start — in that case, quietly say: 'Start with what happens to water in the leaf when the sun is very strong. What are the stomata doing?' With 2 minutes remaining say: 'You should be on Part 3 now. Connect the science to the farmer.' With 30 seconds remaining say: 'Add your name and the date.' Collect written responses. Close the lesson by saying: 'Amina is still watching her sukuma wiki. Now she has your explanation. And so do you.' WAIT TIME: During the 8 minutes of individual writing, maintain a calm, quiet classroom — do not fill the silence with talk. The silence is productive thinking time.	Individual structured writing as a summative sense-making task — requiring learners to translate model-level understanding into linear prose forces precise causal reasoning and accurate vocabulary use. The three-part structure mirrors the phenomenon (wilting, pallor, agricultural application), ensuring the explanation is grounded in the anchoring context rather than abstract recall. The Kenyan context prompt connects biological science to lived experience and future citizenship, reinforcing the AERES values of relevance and responsibility.	The three-part written explanation serves as the primary summative performance task for Sub-strand 2.1. The teacher uses a three-level rubric when marking: Emerging — learner describes observations without causal mechanisms or uses vocabulary incorrectly; Developing — learner provides partial causal chains for one or both phenomena with mostly correct vocabulary; Achieved — learner provides complete, accurate, causally connected explanations for both phenomena and makes a relevant, evidence-based connection to Kenyan agricultural practice. Learners who score Emerging are identified for targeted support in the following term. All learners receive written feedback within one week.

D. TEACHER REFLECTION

After the lesson, consider the following: Did the Evidence Harvest table give learners enough structure to retrieve distributed knowledge, or did some groups simply copy labels from station cards without making causal connections — and if so, how would you revise the table prompt? Did the 12-minute model-building window allow sufficient depth, or did groups rush and produce superficial diagrams — would extending to 15 minutes and reducing the gallery walk improve quality? Which DQB questions remained most difficult to resolve, and what does this reveal about which lessons in the sequence need strengthening in future delivery? Did the three-part written task successfully discriminate between learners who understood at a surface level and those who understood causally — and were the vocabulary constraints

helpful or limiting? Finally, consider whether the Kenyan agricultural context in Part 3 felt genuine and meaningful to learners, or whether a more specific local scenario (e.g., a named county extension officer, a KALRO recommendation, or a specific cooperative) would make the connection more authentic next time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We revisited the original sukuma wiki wilting and pallor phenomenon and reviewed evidence gathered across nine lessons through annotated predictions, an Evidence Harvest gallery walk, and group annotated models integrating leaf structure, photosynthesis stages, limiting factors, and mineral nutrition.
What did I learn?	The wilting of sukuma wiki at midday — even when soil is wet — occurs because high light intensity and temperature accelerate transpiration through open stomata faster than roots can absorb and transport water upward through xylem, causing loss of turgor pressure in leaf cells; this is a reversible process that reverses at night when transpiration slows and turgor is restored. The pallor and slow growth in shade occur because reduced light intensity limits the light-dependent reactions, reducing ATP and NADPH production, which in turn limits the Calvin cycle output of glucose; with less glucose available for growth and less chlorophyll synthesised (especially if nitrogen or magnesium are also limiting), the plant grows slowly and appears pale yellow-green.
How does this explain the phenomenon?	By integrating leaf anatomy, the two-stage mechanism of photosynthesis, the concept of limiting factors, and the roles of soil minerals, we can fully resolve the driving question: the sukuma wiki plant needs light as an energy source for the light-dependent reactions, water and CO ₂ as raw materials for both stages, chlorophyll to absorb light, and minerals such as nitrogen and magnesium from the soil to build chlorophyll and enzymes — when any of these factors is limiting, the plant cannot make food efficiently, leading to the observable symptoms Amina saw in her Nairobi kitchen garden; understanding these mechanisms helps Kenyan smallholder farmers make evidence-based decisions about irrigation timing, shade management, and fertiliser application to improve sukuma wiki yield and contribute to household and national food security.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.